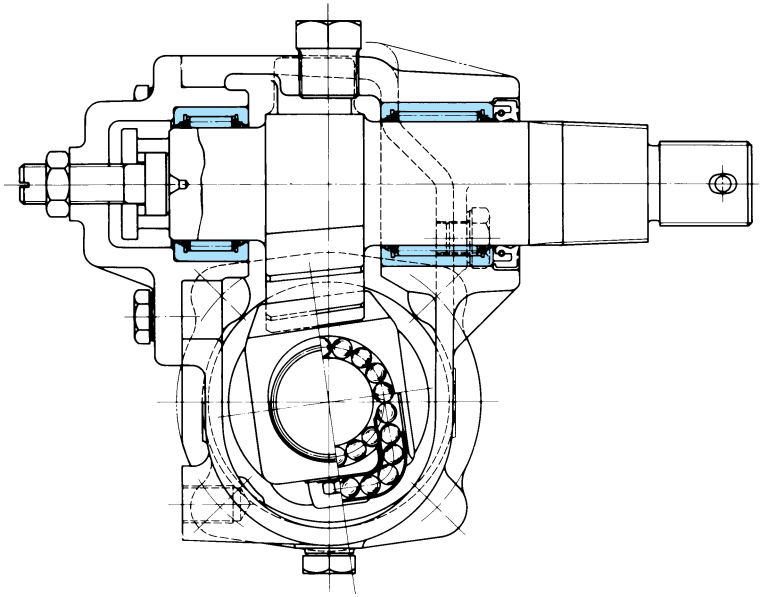


Fig. 5 Ball Screw Steering Sector Shaft

For large trucks and buses, solid bearings are used. For passenger cars, however, drawn cup needle roller bearings are applied in many cases.

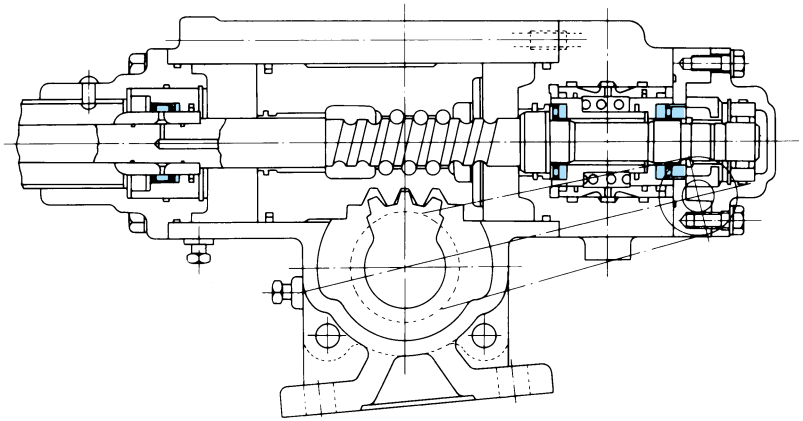


Fig. 6 Power Steering Systems

Drawn cup needle roller bearings and thrust needle roller bearings are applied to the hydraulic mechanism in power steering systems.

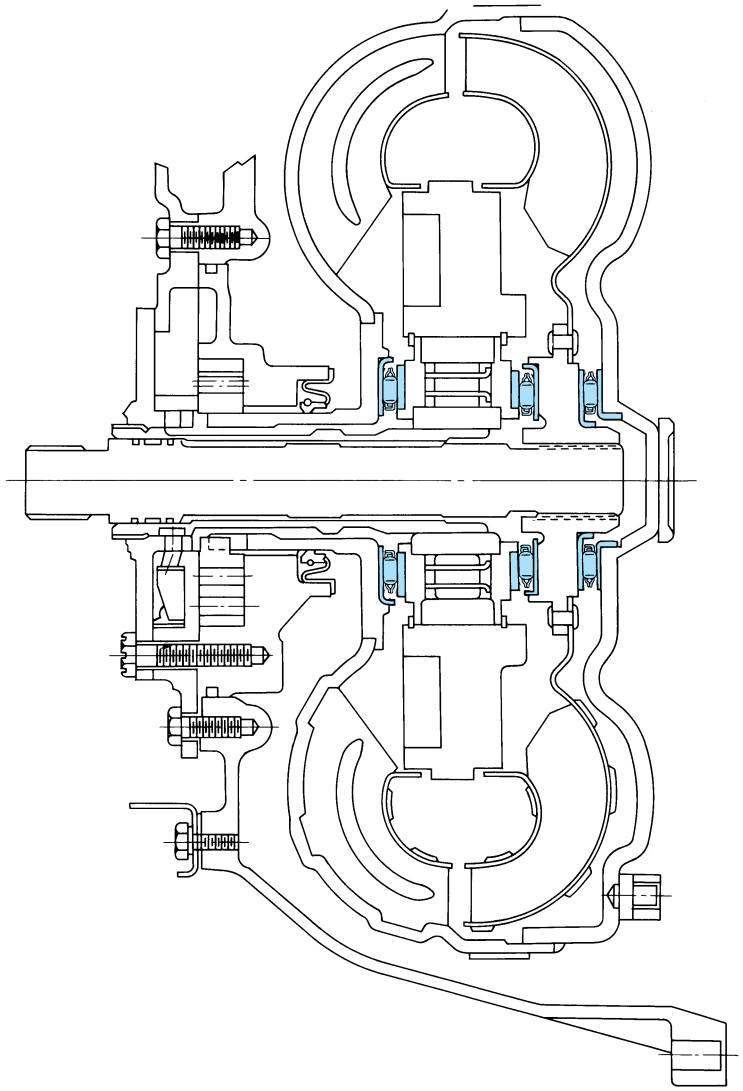


Fig. 7 Torque Converter

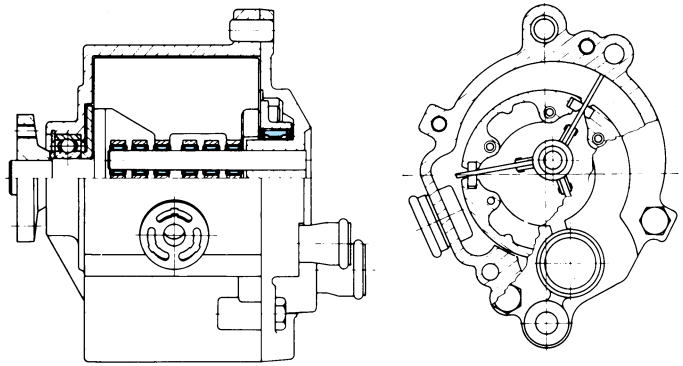


Fig. 8 Air Pump (3-vane)

In order to reduce size and weight, drawn cup needle roller bearings with a cage are applied to the vanes and rotor shaft. A two-vane type is also available.

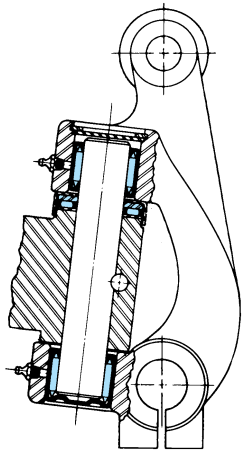


Fig. 9 Kingpin

Drawn cup needle roller bearings are applied to the kingpin. In some cases, a grease nipple is attached for lubrication.

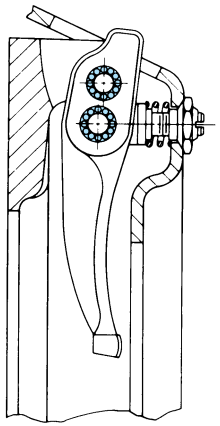


Fig. 10 Clutch Finger

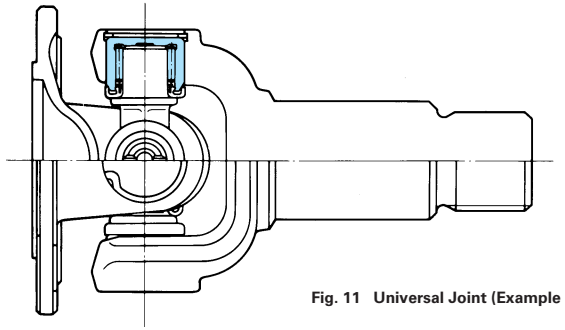


Fig. 11 Universal Joint (Example of Solid Bearing Application)

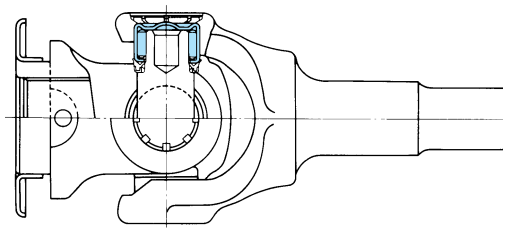


Fig. 13 Universal Joint (Example of Drawn Cup Bearing Application)

This illustrates an application of drawn cup needle roller bearings in a universal joint to reduce size, weight and cost.

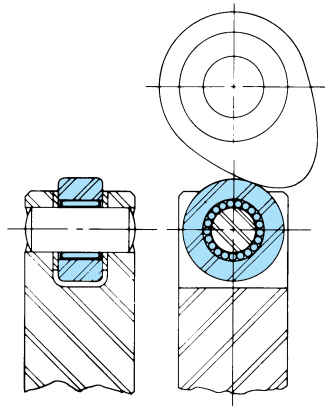


Fig. 12 Engine Camshaft Follower

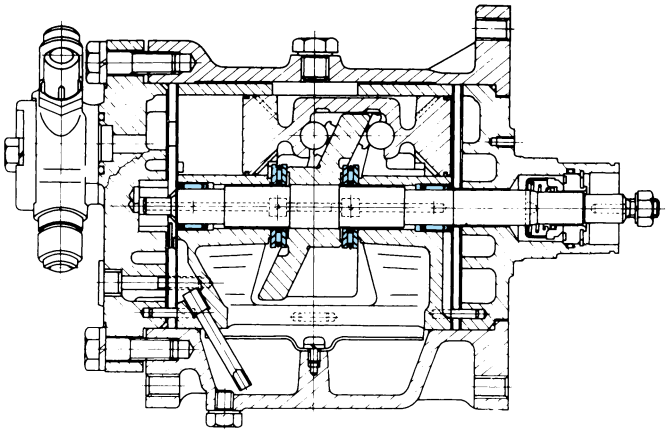


Fig. 14 Automotive Air-Conditioner Compressor
This illustrates the application in the compressor of a swash plate air conditioner, with a characteristic structure involving support by a thrust bearing.

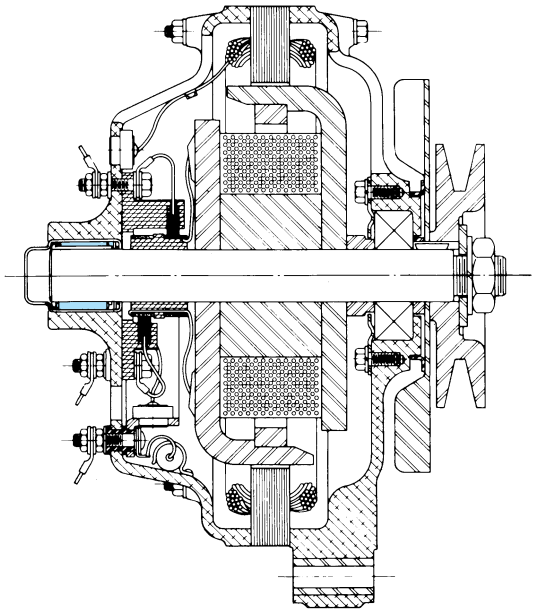


Fig. 15 Alternator (AC Generator)
A drawn cup needle roller bearing with a cage is applied on the opposite side of the pulley to reduce size, weight and cost.

CONSTRUCTION AND CARGO HANDLING EQUIPMENT

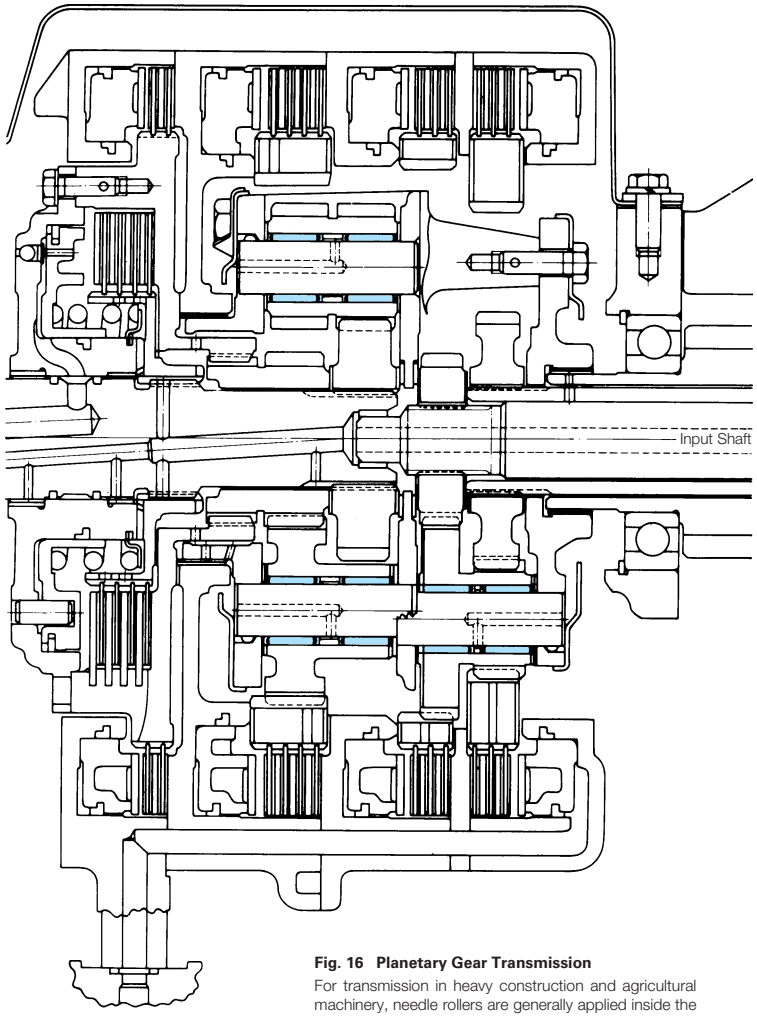


Fig. 16 Planetary Gear Transmission
For transmission in heavy construction and agricultural machinery, needle rollers are generally applied inside the planetary gear and placed in two rows for stability. For assembly reasons, cage & needle roller assemblies are also used.

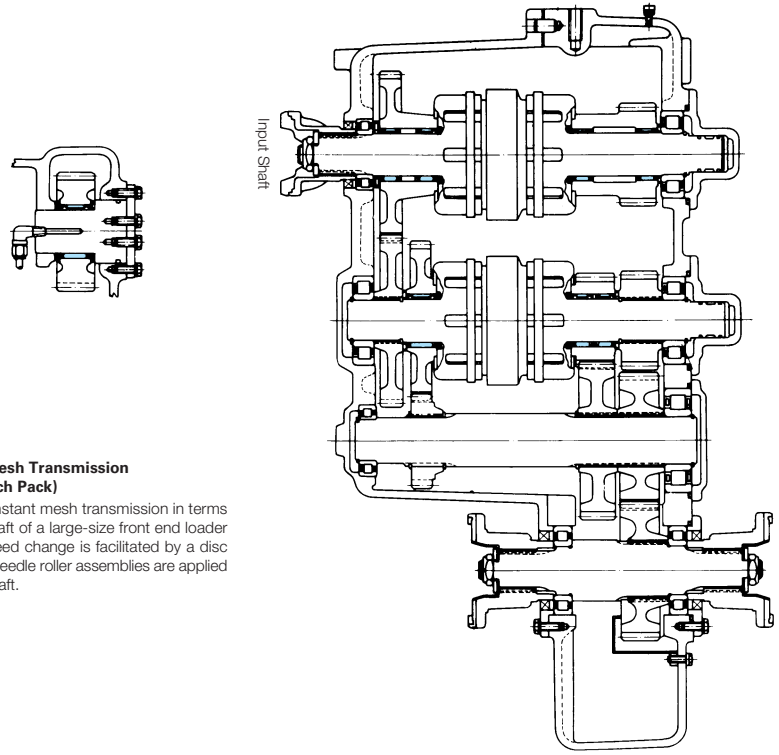


Fig. 17 Constant Mesh Transmission (Using Clutch Pack)

This illustrates the constant mesh transmission in terms of the intermediate shaft of a large-size front end loader with rubber tires. Speed change is facilitated by a disc clutch pack. Cage & needle roller assemblies are applied to this intermediate shaft.

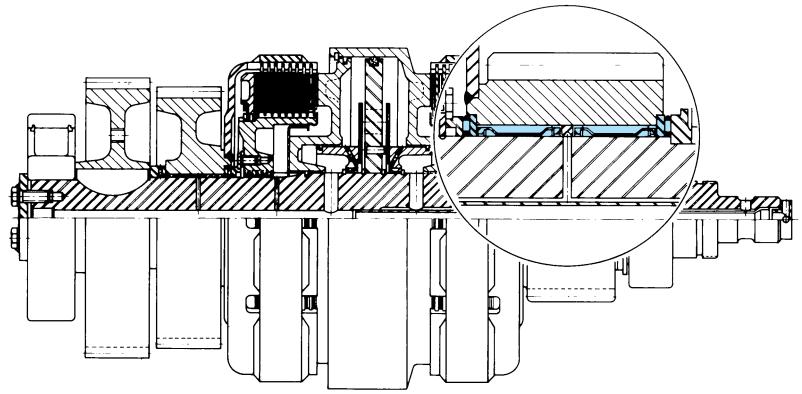


Fig. 18 Detailed Drawing of Clutch Pack

Two units of cage & needle roller assemblies are commonly applied to the gear support of the clutch pack. This bearing usually runs with a light load or without any load and does not run if a load is imposed.
In such an application, a bearing with a static load rating of 4 to 7 times that of the static load should be selected.

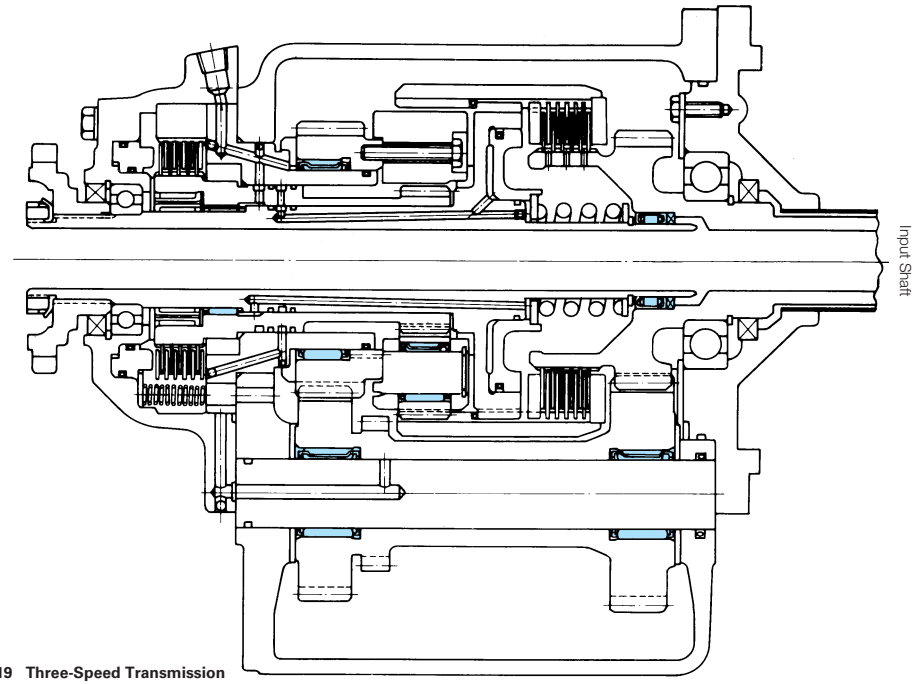


Fig. 19 Three-Speed Transmission

The figure shows an example of the three-speed transmission of a garden tractor or the like. Cage & needle roller assemblies are applied to the countershaft and planetary gear. Cage & needle roller assemblies are also used as clutch support on the right-hand side of a one-way clutch.

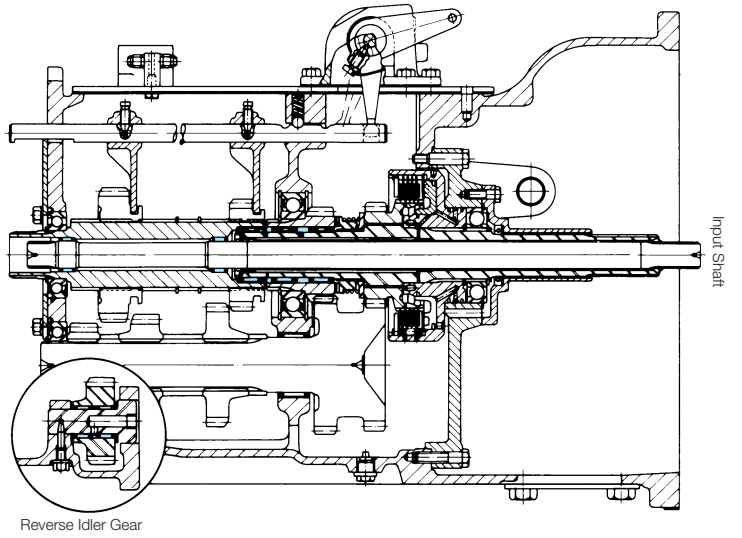


Fig. 20 Six-Speed Transmission

The figure shows a six-speed sliding mesh transmission, which is basically a three-speed transmission with a two-speed power shift in the earlier stage.

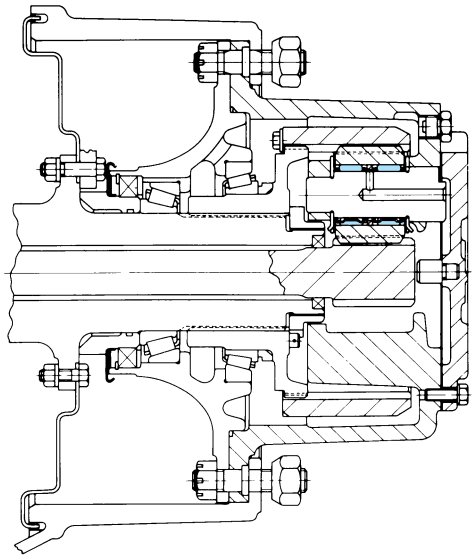


Fig. 21 Final Reduction Gear for Large Tractor

For rubber-tire tractors used in construction, a planetary final reduction gear is used as shown in the figure. In this case, since the load is large and space is limited, cage & needle roller assemblies are used in consideration of the large load capacity, compact size, assembly, and service.

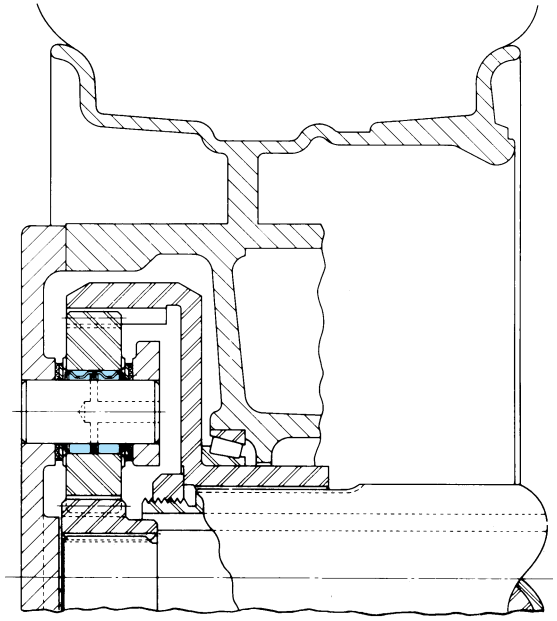


Fig. 22 Final Reduction Gear

The structure is similar to **Fig. 21**. Two rows of cage & needle roller assemblies are also applied to this device to enhance the stability of the gears. Rollers treated with a proper crowning are recommended.

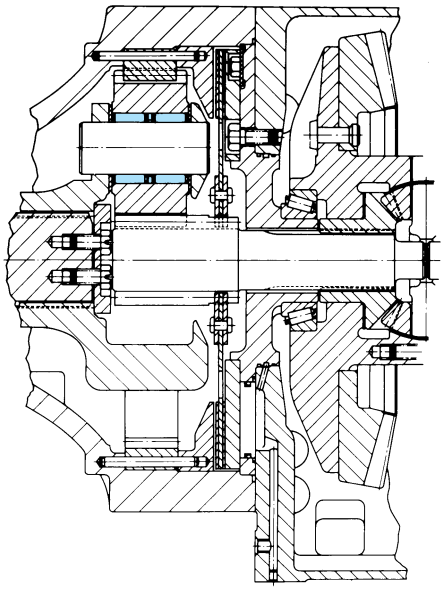


Fig. 23 Final Reduction Gear

In some cases, a final reduction gear is positioned close to the differential gears as shown in this figure. Instead of cage & needle roller assemblies, a single needle roller bearing is integrated into the two rows and the rollers are kept separate by a hardened spacer. It is absolutely essential that the needle rollers have the proper crowning.

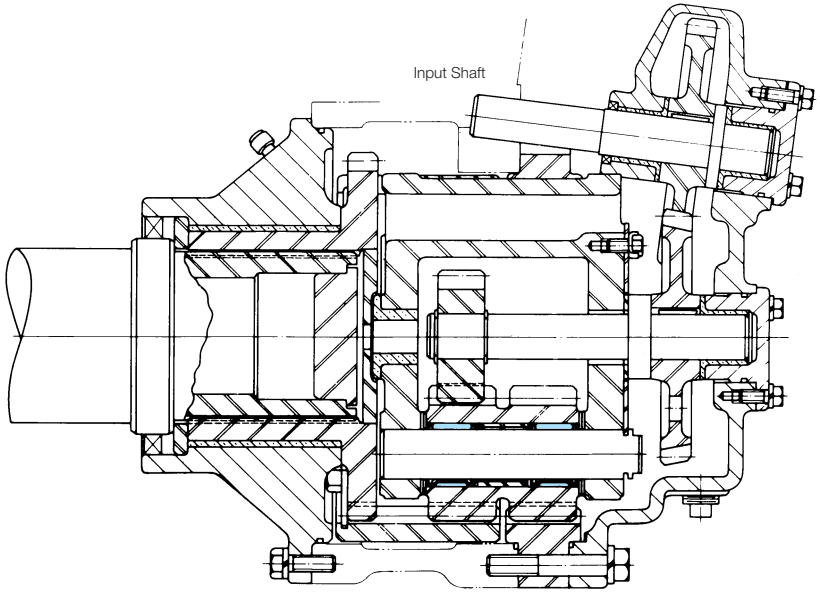


Fig. 24 Auxiliary Gearbox

This illustrates an application of cage & needle roller assemblies to the planetary gears of an auxiliary gearbox.

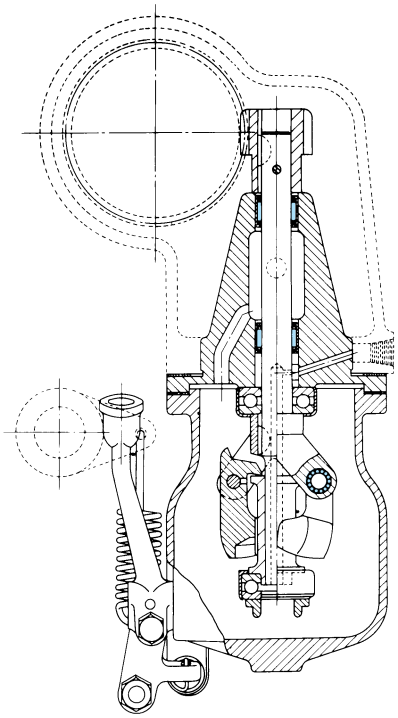


Fig. 25 Governor

The governor of a diesel engine is one of the parts where fretting corrosion or false brinelling is easily caused by vibration. In the figure, a drawn cup, full complement bearing is applied to the pivot arm.

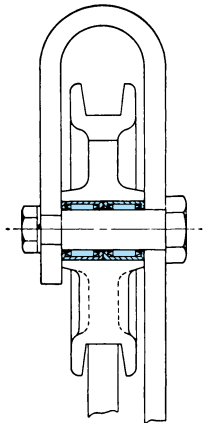


Fig. 26 Sheave for Cargo Handling Vehicle

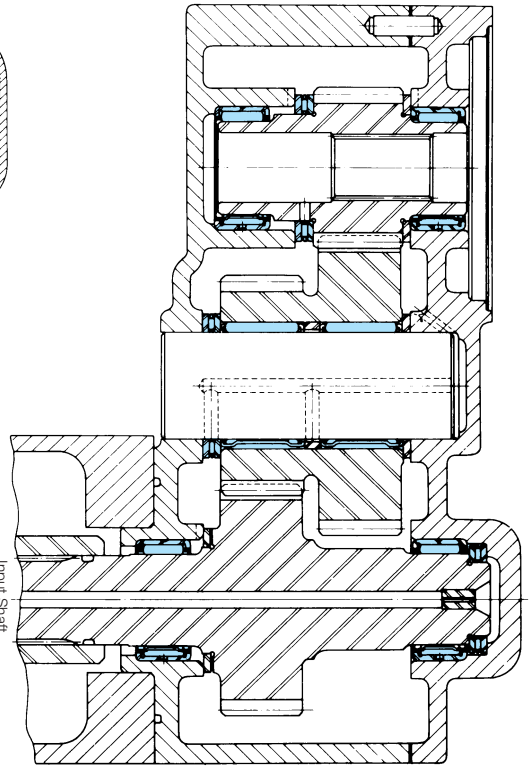


Fig. 27 Hydraulic Pump Drive Unit

For hydraulic pump drive units in equipment such as large scrapers, which generally use aluminum or copper alloy metal, if forced lubrication is not possible, the application of a needle roller bearing facilitates lubrication as shown in the figure.

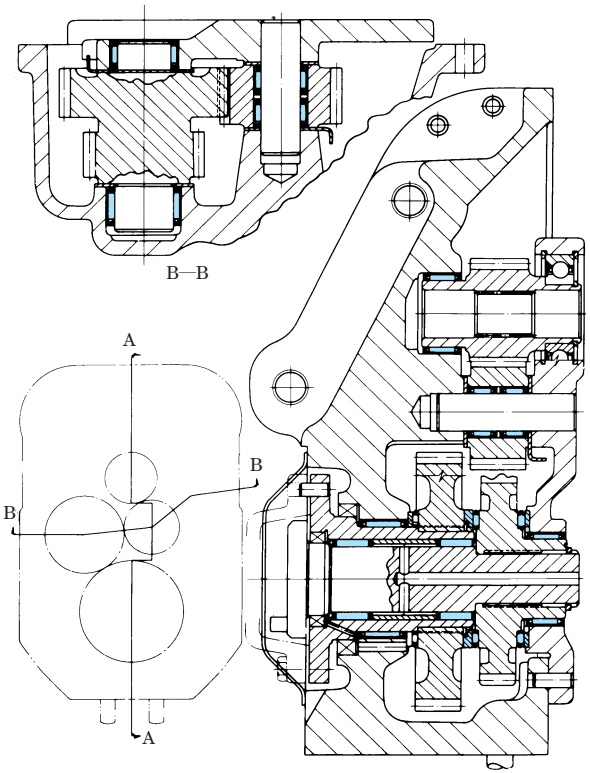


Fig. 28 Two-Speed Transmission PTO Unit

The power take-off (PTO) of a large two-speed tractor for agricultural use is successfully designed very compactly with a combination of drawn cup needle roller bearings and thrust bearings.

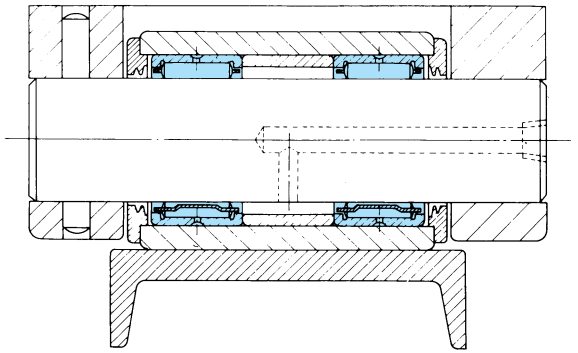


Fig. 30 Forklift and Truck Roller

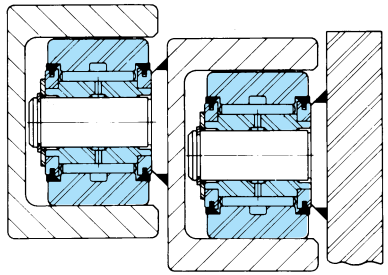


Fig. 29 Truck Roller in Forklifts

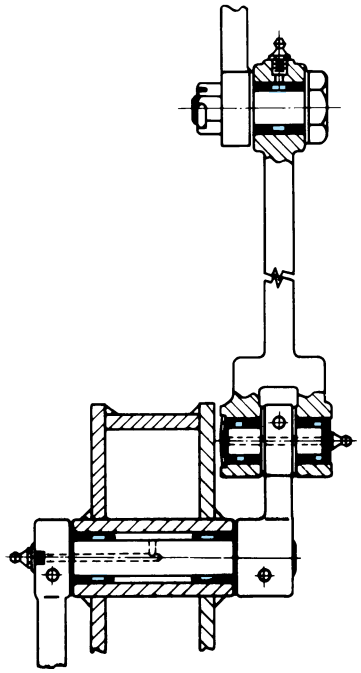


Fig. 31 Control System Linkage

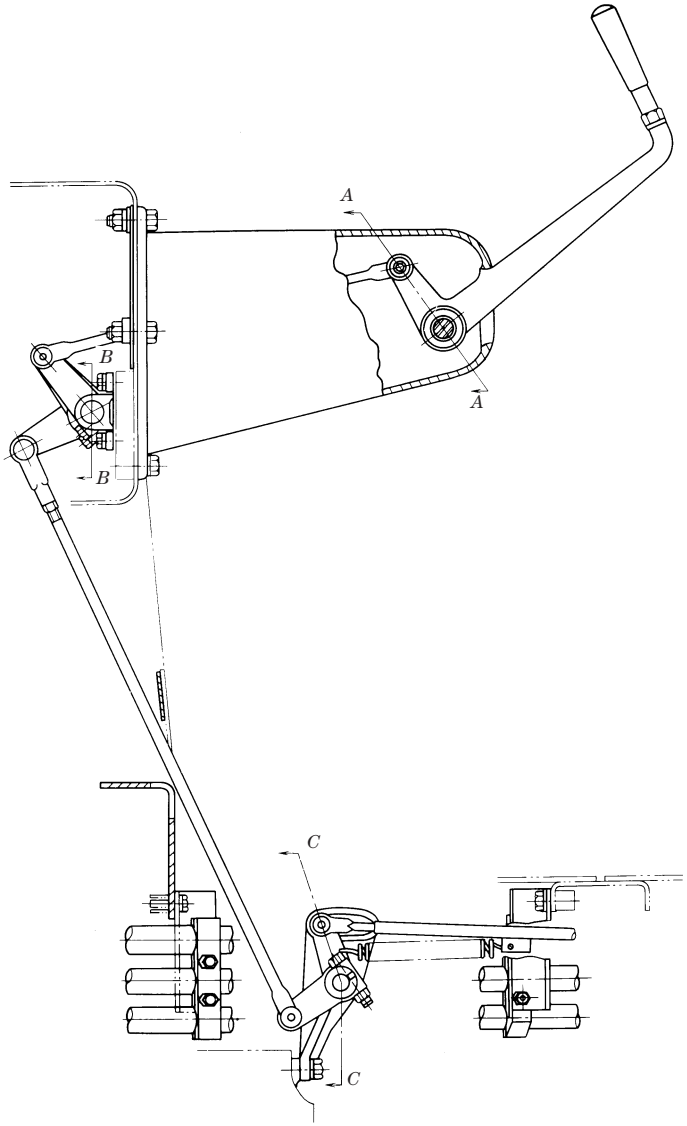


Fig. 32 Control Lever System

For the control lever system in equipment such as construction machinery, there has been a switch from plain metal to needle roller bearings for smoother performance and lubrication. Depending on the applicable part, due consideration is sought for enabling re-lubrication. Detailed cross-sectional drawings for each part are shown on the following page.

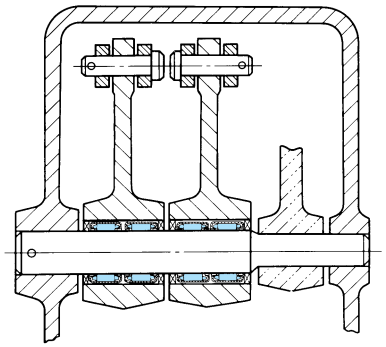


Fig. 32.1 Cross-Section A-A

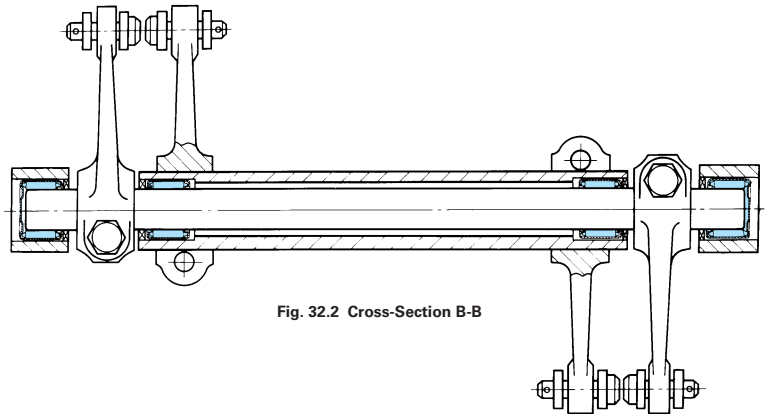


Fig. 32.2 Cross-Section B-B

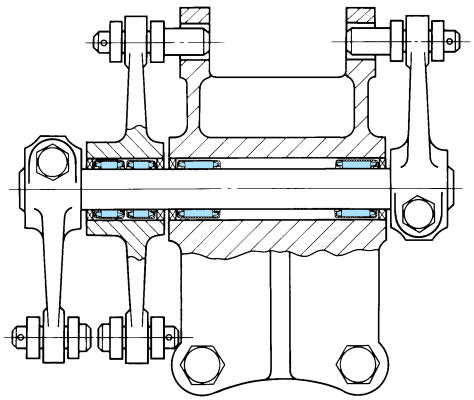


Fig. 32.3 Cross-Section C-C

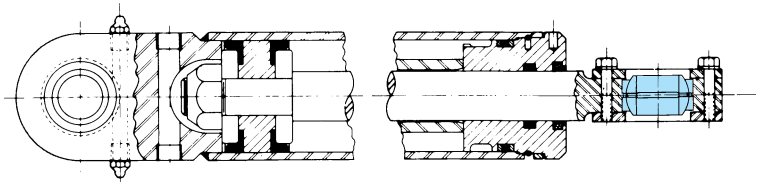


Fig. 33 Hydraulic Cylinder for Bulldozer Blade Operation

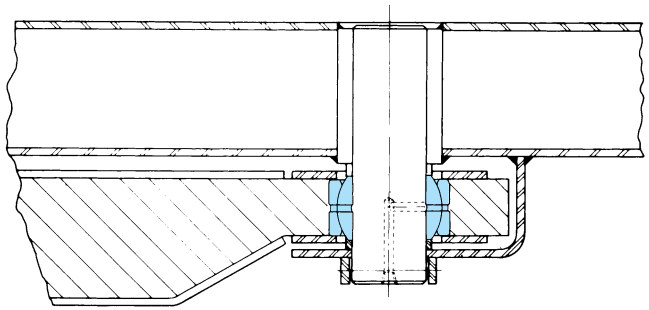


Fig. 34 Scraper Linkage

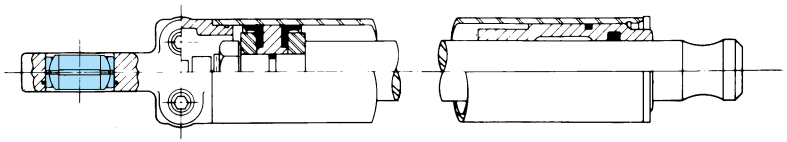


Fig. 35 Hydraulic Cylinder for Shovel Operation

AGRICULTURAL MACHINERY

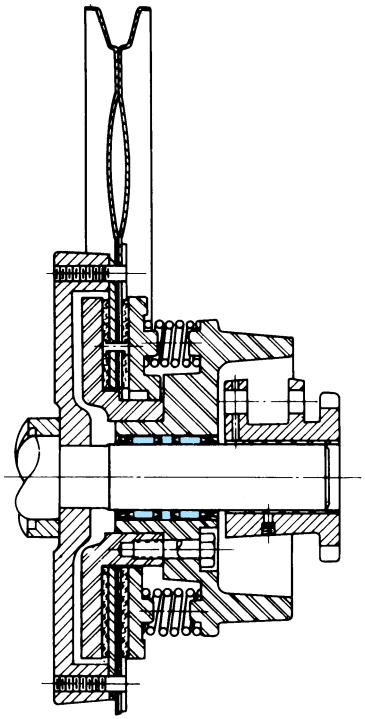


Fig. 36 Grain Reaping Machine

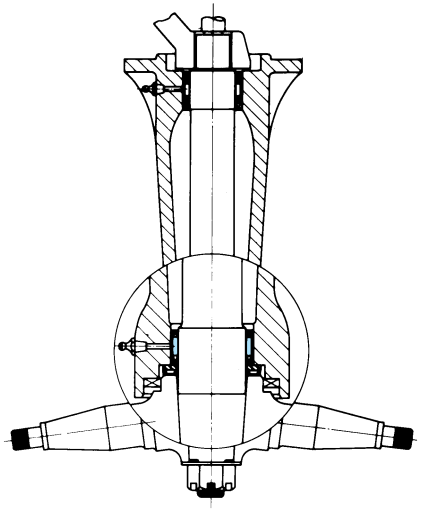


Fig. 37 Tractor Bolster
A compact thrust needle roller bearing is applied to the thrust pad of a tractor bolster. A drawn cup needle roller bearing is also used at the same time for smoother operation.

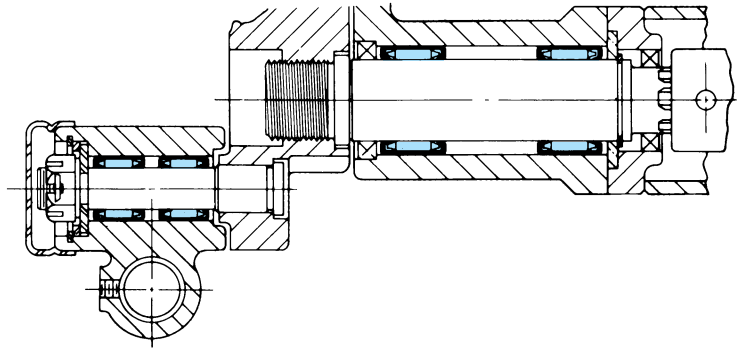


Fig. 38 Pitman
For superior wear resistance and operating efficiency, a drawn cup needle roller bearing is applied to the pitman crank and crankpin.

MACHINE TOOLS

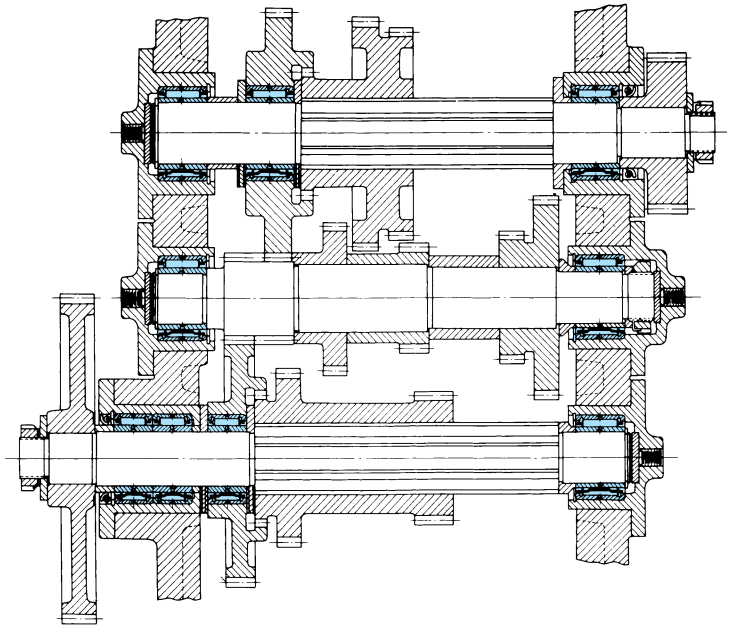


Fig. 39 Machine Tool Gearbox

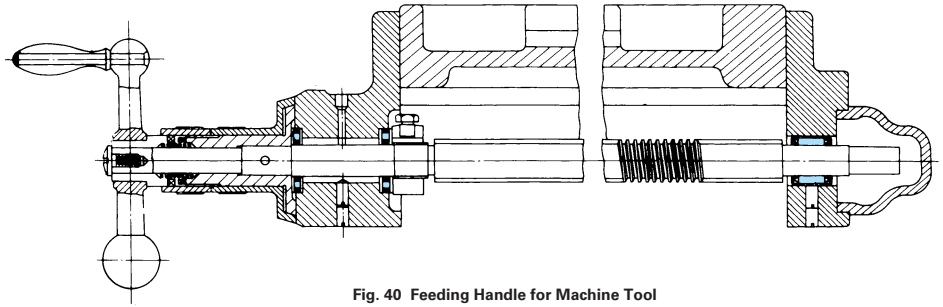


Fig. 40 Feeding Handle for Machine Tool

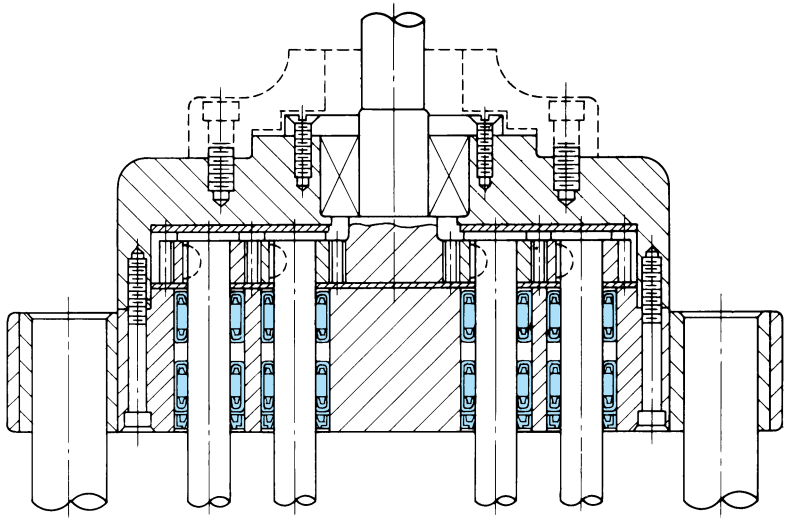


Fig. 41 Multi-Spindle Drill

In the case of multiple spindles occupying a limited space, such as a multiple spindle drilling machine, a drawn cup needle roller bearing is recommended.

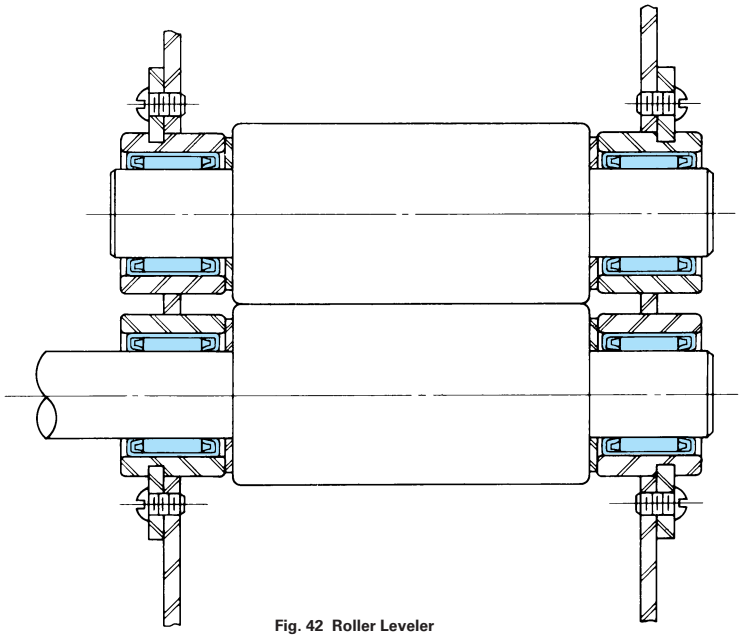


Fig. 42 Roller Leveler

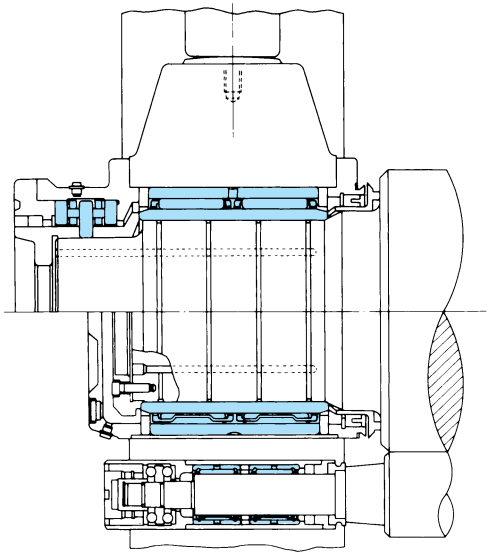


Fig. 43 Back-Up Roll and Working Roll

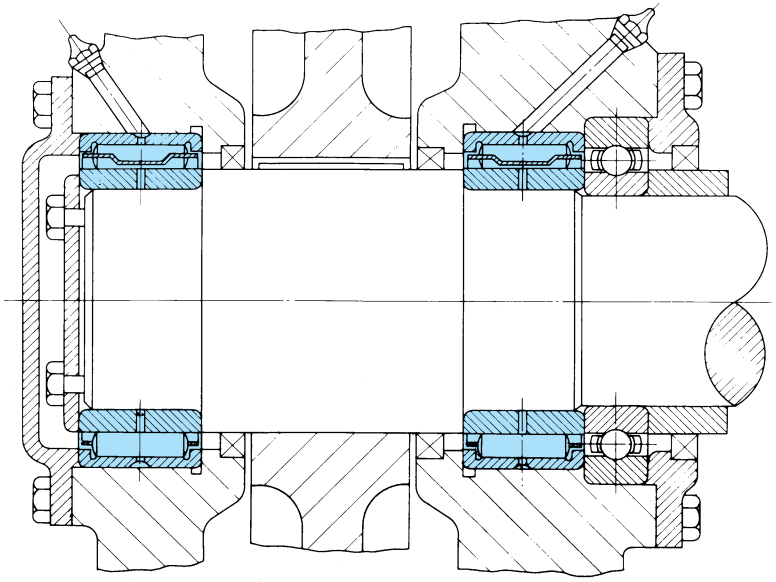


Fig. 44 Crane Wheel

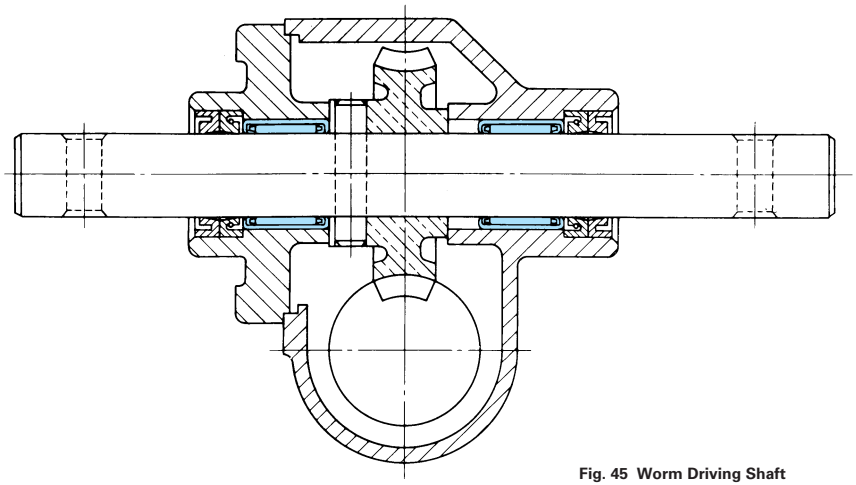


Fig. 45 Worm Driving Shaft

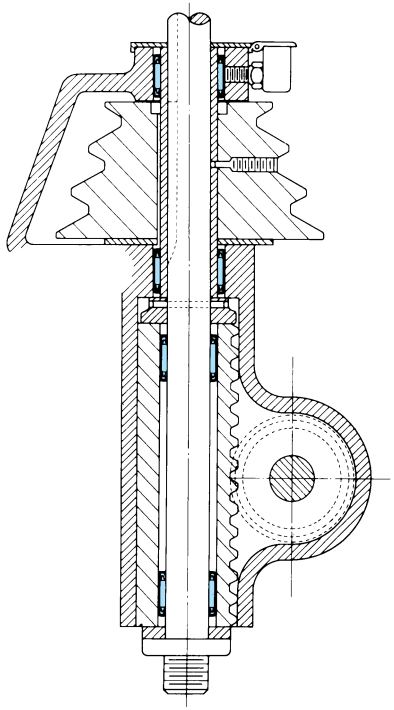


Fig. 46 Small Drill Press Spindle

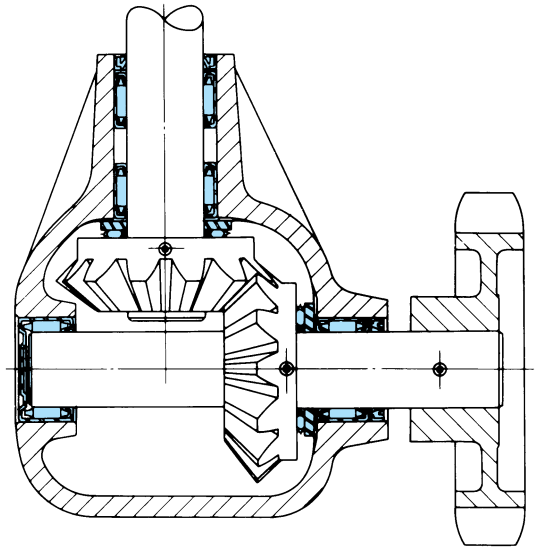


Fig. 47 Gearbox

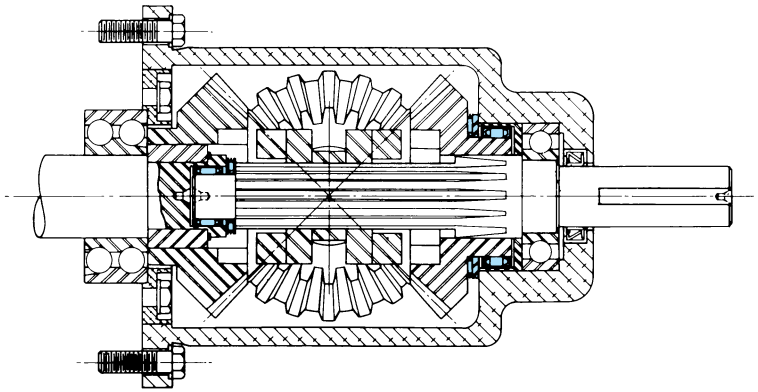


Fig. 48 Gearbox

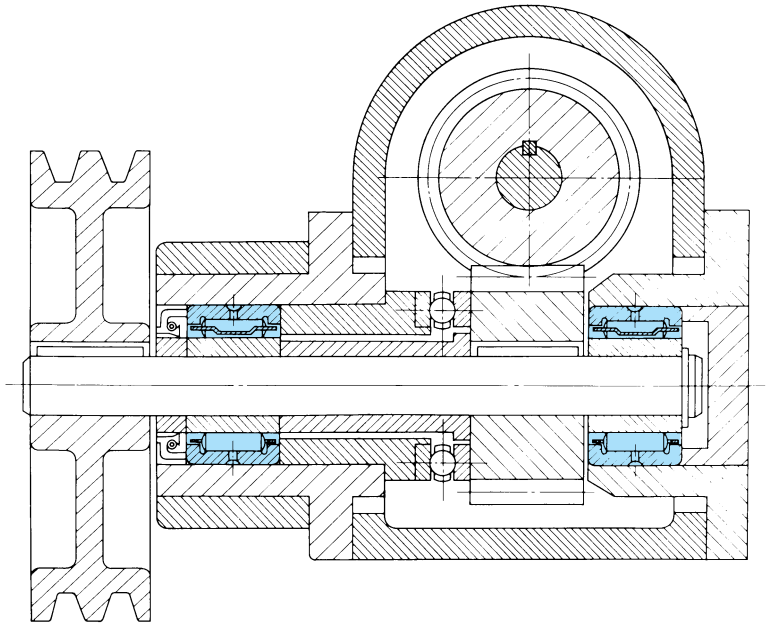


Fig. 49 Conveyer Driving Reduction Gear

BUSINESS MACHINES

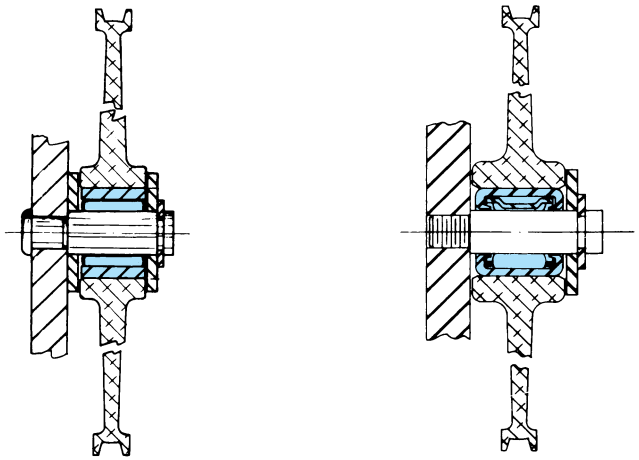


Fig. 50 Magnetic Tape Driving Pulley for Electronic Computers

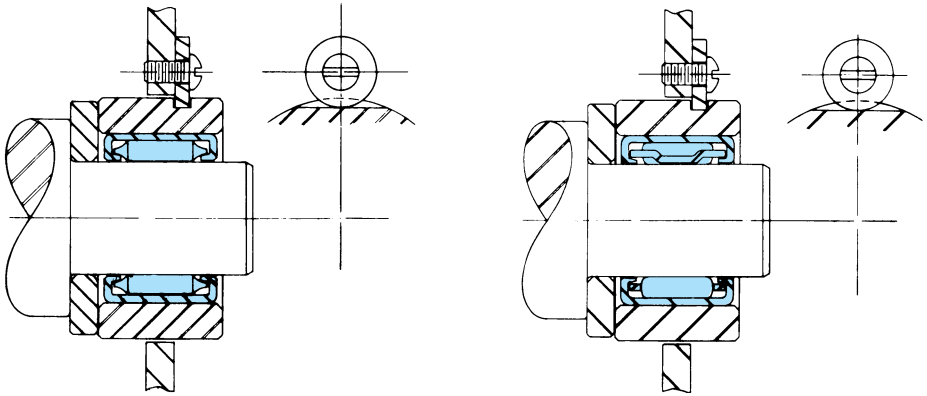


Fig. 51 Copying Machine Side Frame

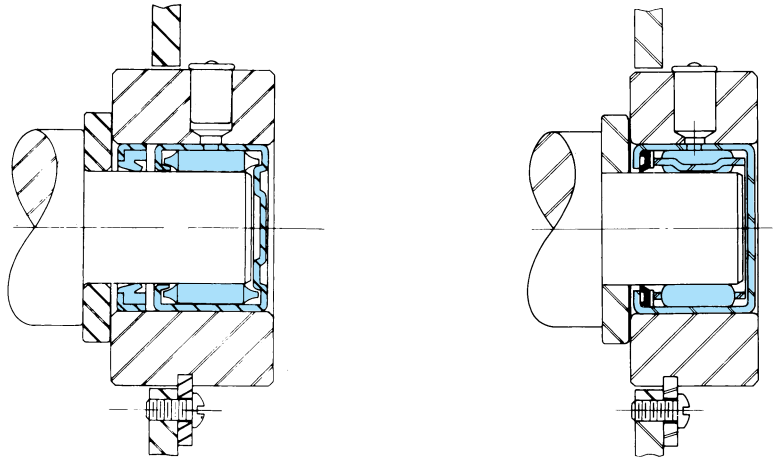


Fig. 52 Copying Machine Side Frame

The application of a closed drawn cup needle roller bearing to the shaft end contributes to cost reduction and enhancement of seal performance.

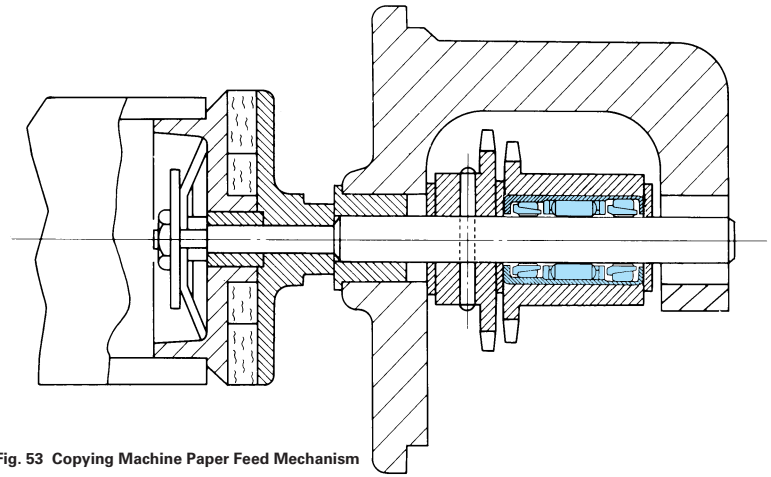


Fig. 53 Copying Machine Paper Feed Mechanism

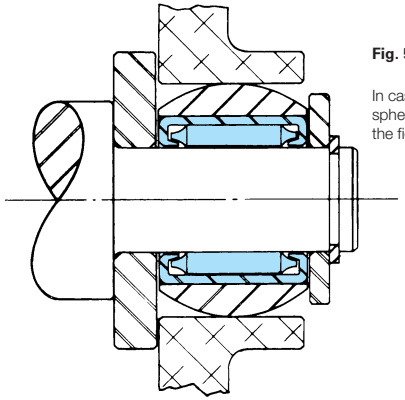


Fig. 54 Copying Machine Side Frame (In Case of Frequent Misalignment)

In case misalignment cannot be avoided, the spherical housing may be used as shown in the figure.

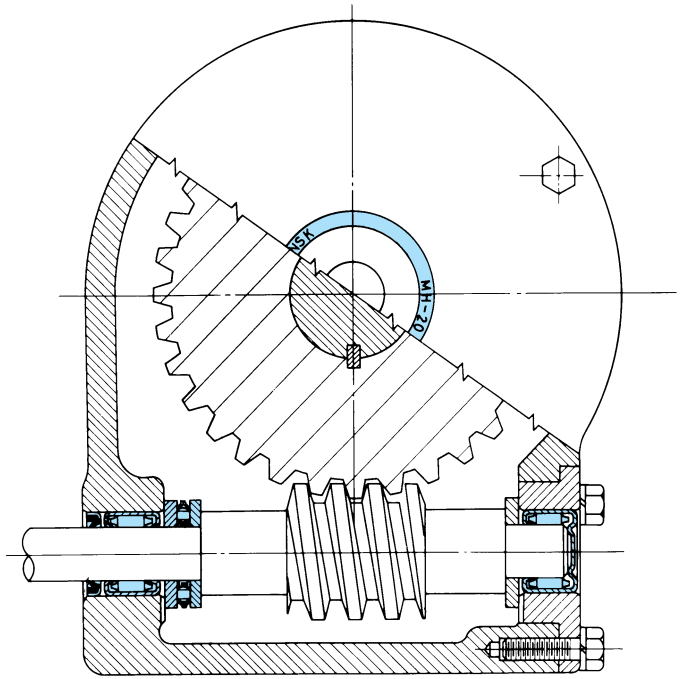


Fig. 55 Speed Reducer for Business Machines

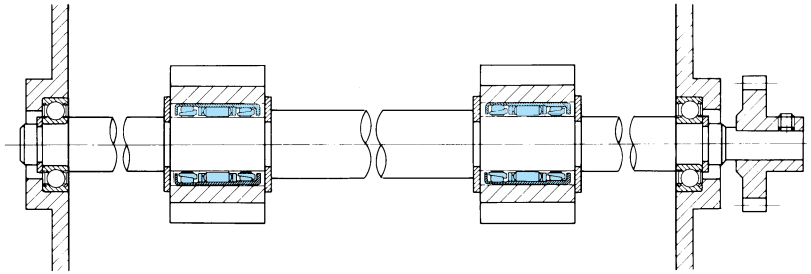


Fig. 56 Copying Machine Paper Feed Roll

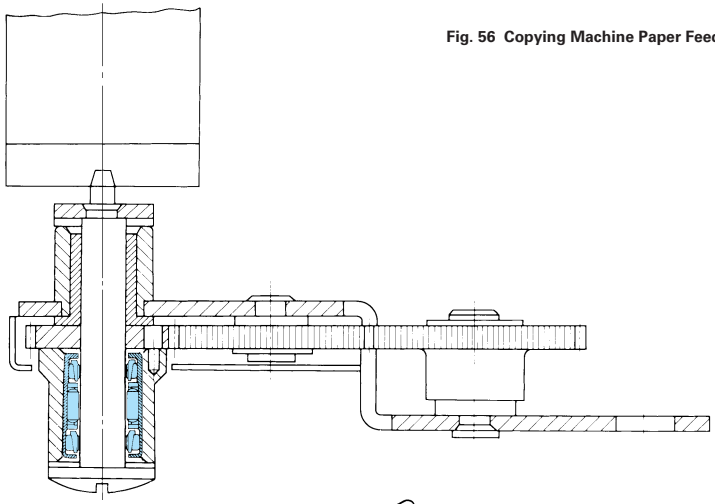


Fig. 57 Copying Machine Paper Feed Roll Mechanism

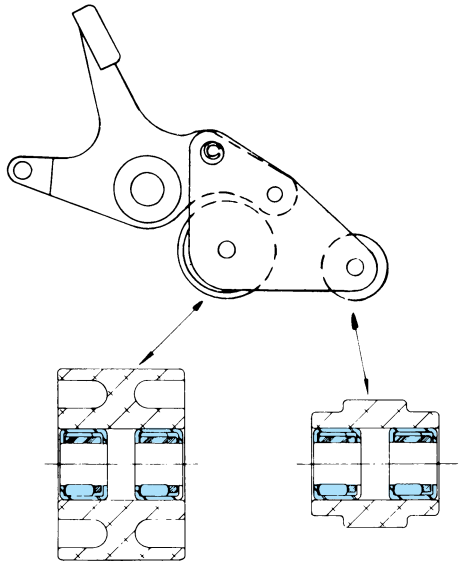


Fig. 58 Copying Machine Drive Roll (With Small Roll Diameter)

ELECTRIC AND PNEUMATIC EQUIPMENT

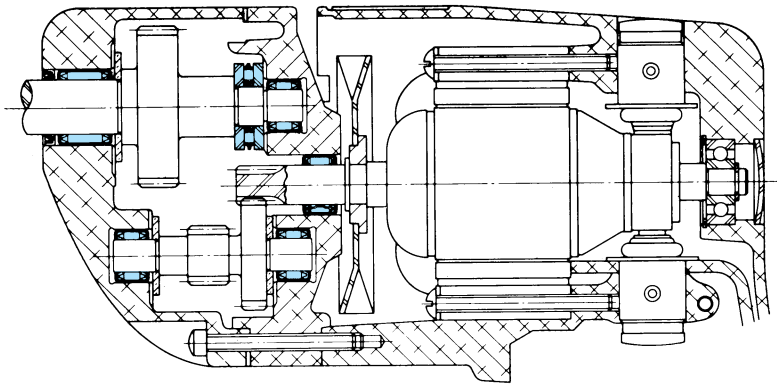


Fig. 59 Electric Drill

The application of a drawn cup needle roller bearing contributes to shortening the distance between shafts and reduces size and weight.
A needle roller bearing with a cage is applied to the motor shaft due to high-speed rotation.

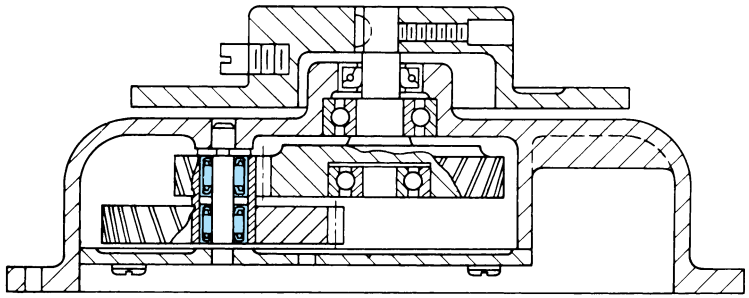


Fig. 60 Electric Polisher

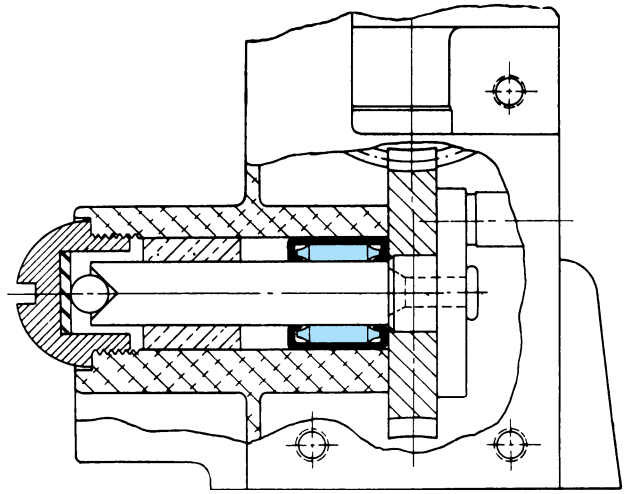


Fig. 61 Hedge Trimmer

The figure shows the crankshaft of a hedge trimmer. A hardened steel ball is used as the center shaft because an axial load through worm gears is imposed there.

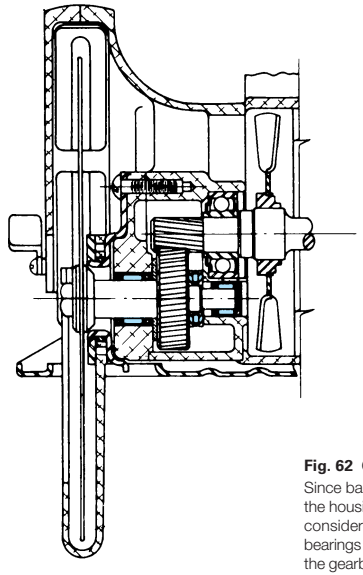


Fig. 62 Circular Saw

Since ball bearings are commonly applied to motor fan shafts, the housing space on the right-hand side of the spindle shaft is considerably limited. Accordingly, drawn cup needle roller bearings are used in many cases. In order to prevent grease in the gearbox from entering into the motor, a closed type bearing is applied.

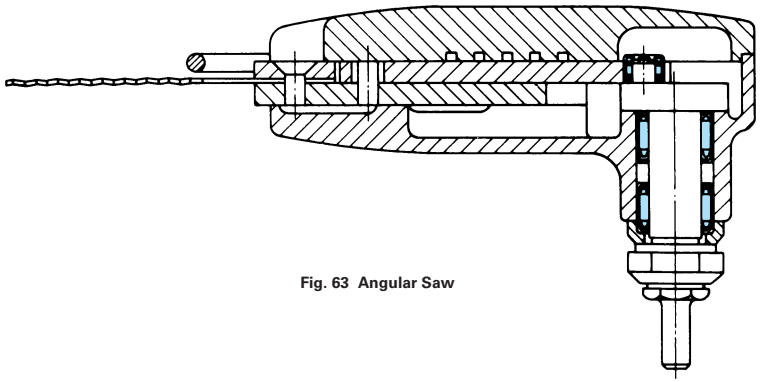


Fig. 63 Angular Saw

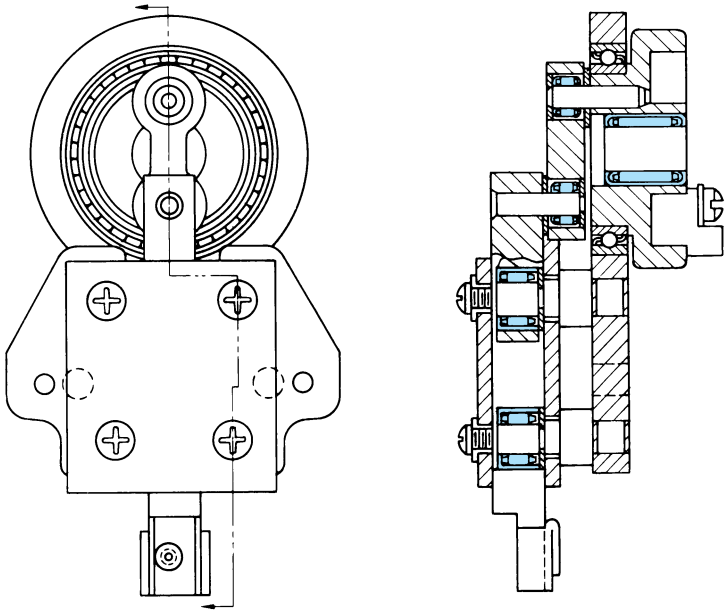


Fig. 64 Shaper Saw

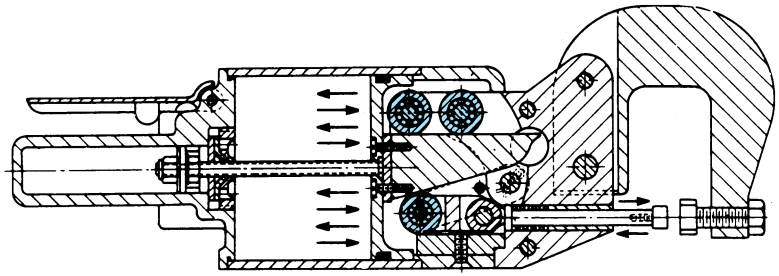


Fig. 65 Rivetter

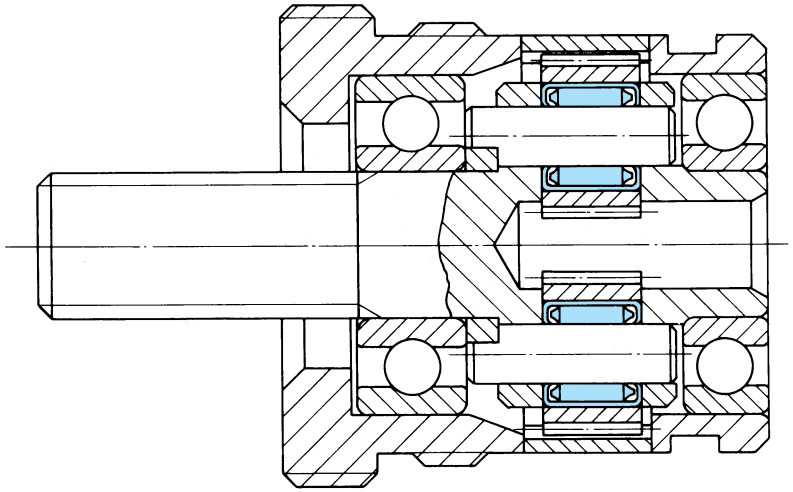


Fig. 66 Planetary Gears of Air Drill

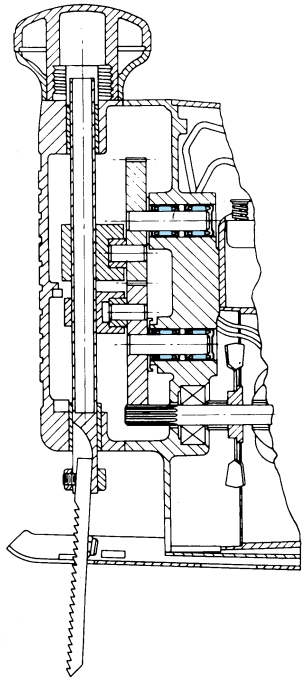


Fig. 67 Jig Saw
Two drawn cup needle roller bearings are applied to support the blade and counter balance driving gears.

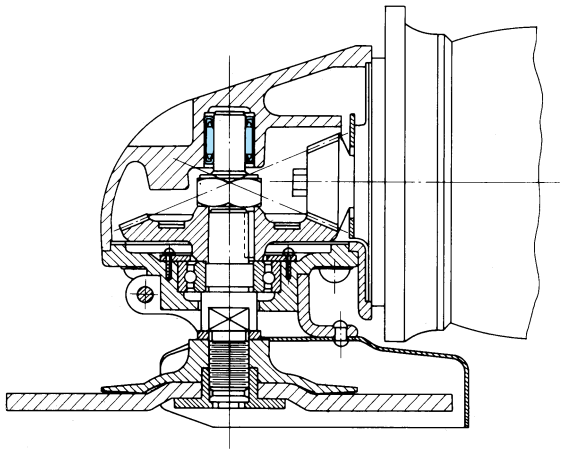


Fig. 68 Polisher
It is possible to use the housing as a through hole by changing the drawn cup needle roller bearing on the upper part of the spindle shaft to a closed type.

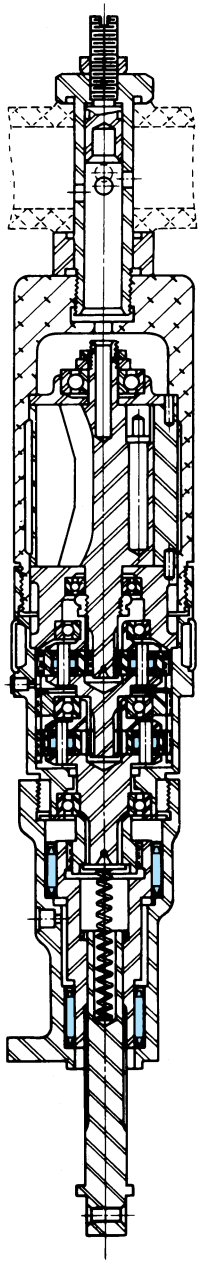


Fig. 69 Nut Setter

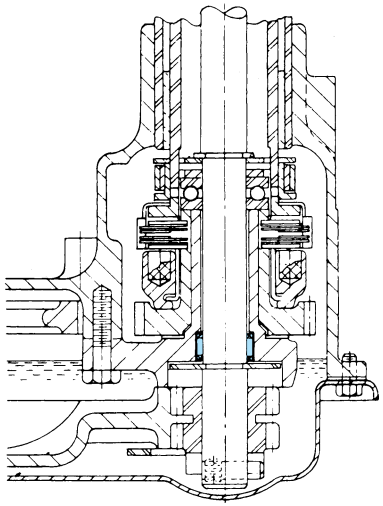


Fig. 70 Agitator Shaft

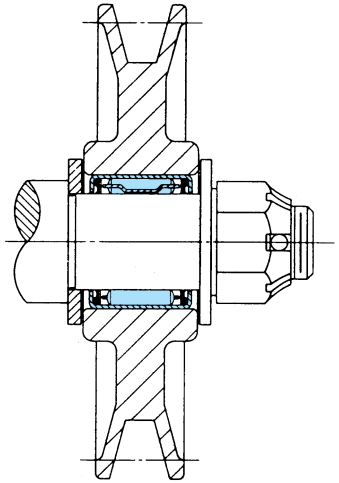


Fig. 71 Pulley Support
This is a good example of low-cost design, showing that a single drawn cup needle roller bearing with a seal can deliver stable performance without maintenance over a long period of time to facilitate long life.

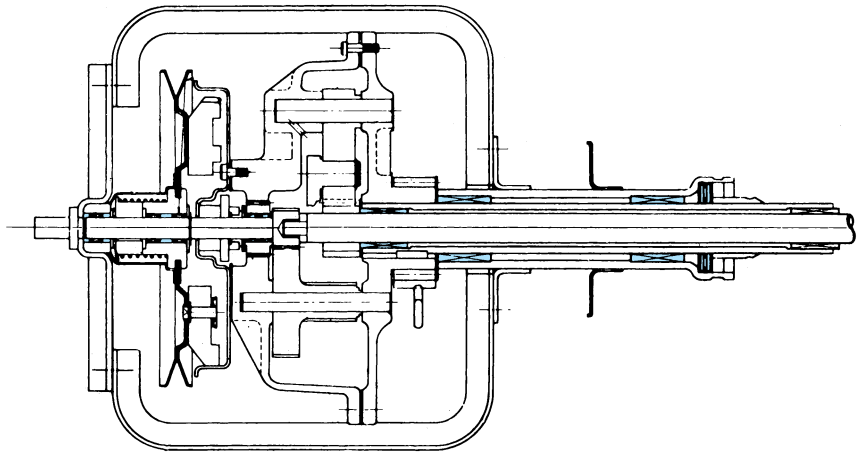


Fig. 72 Agitator Shaft

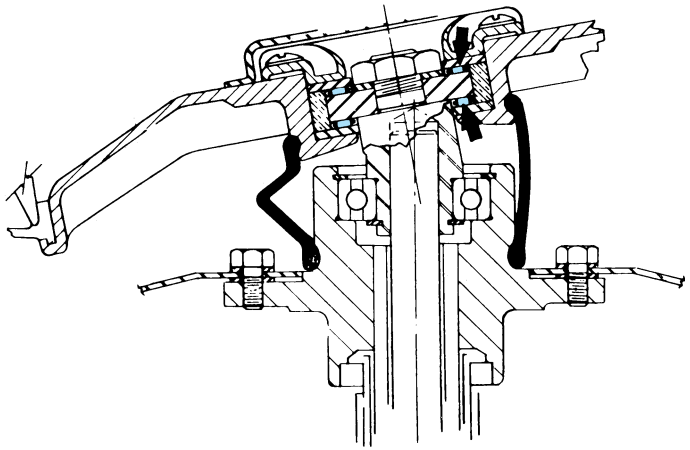


Fig. 73 Spank Mechanism

This is a typical example of a vertical agitator in which a ball bearing was replaced with a needle roller bearing to reduce size.

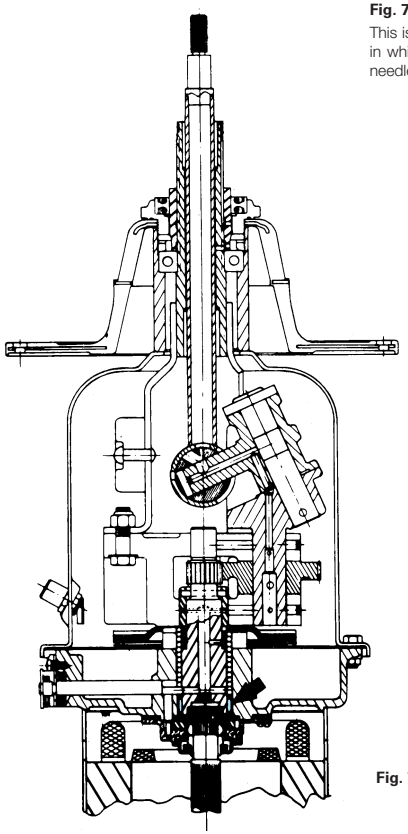


Fig. 74 Spank Agitator

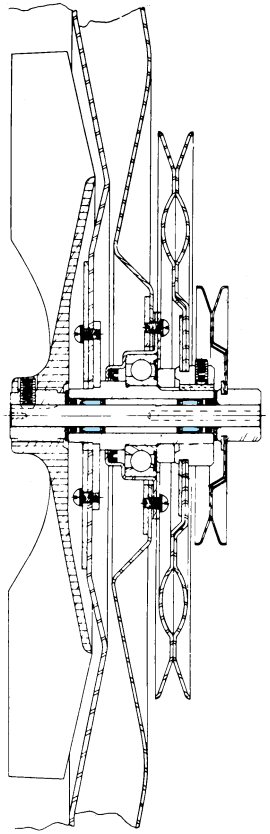


Fig. 75 Washer and Dryer Fan-Drive Pulley

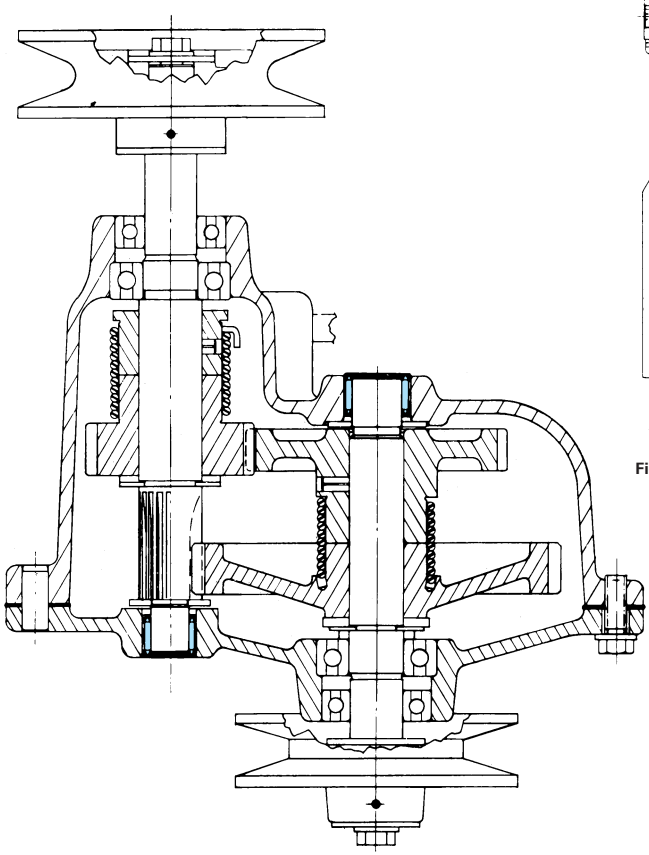


Fig. 76 Washer and Dryer Two-Speed Transmission

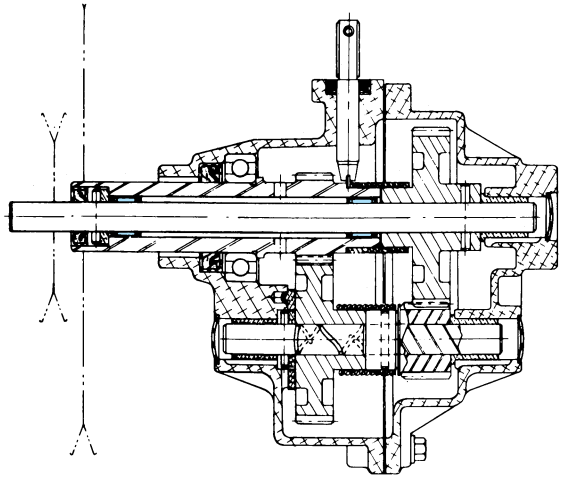


Fig. 77 Washer and Dryer Two-Speed Transmission

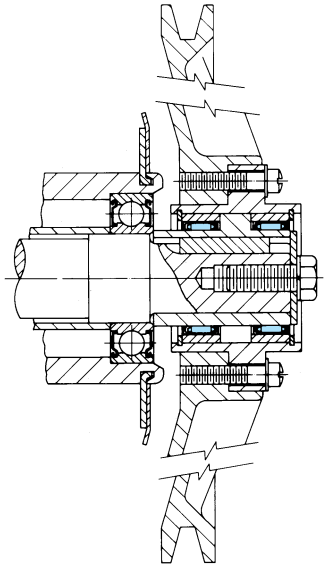


Fig. 78 Washing Machine Basket Drive Shaft

TWO-STROKE ENGINES

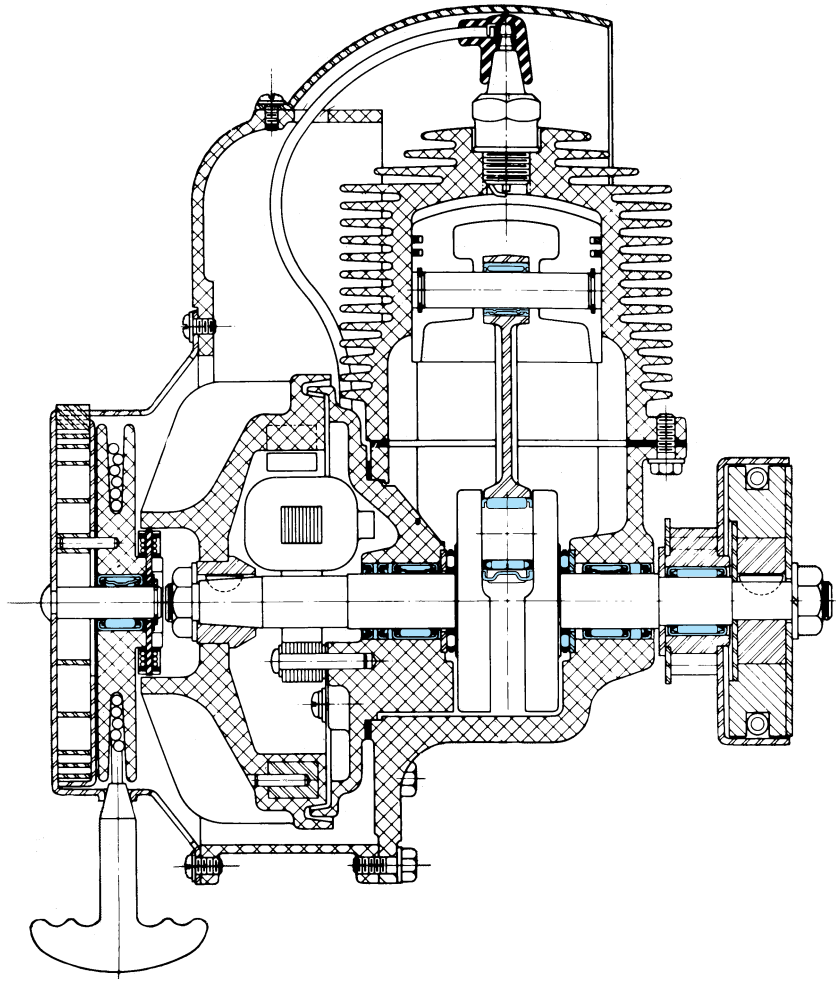


Fig. 79 Chain Saw Engine

Many drawn cup needle roller bearings are applied to the chain saw engine, contributing to its lighter weight. Application of a drawn cup roller clutch to the starter allows further simplification of operation.

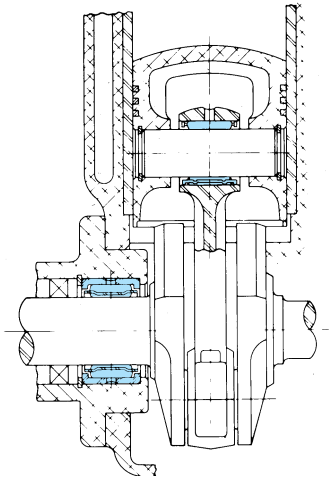


Fig. 80 Two-Stroke Engine
This illustrates the application of a solid needle roller bearing to the crankshaft, increasing load capacity and facilitating a more compact size.

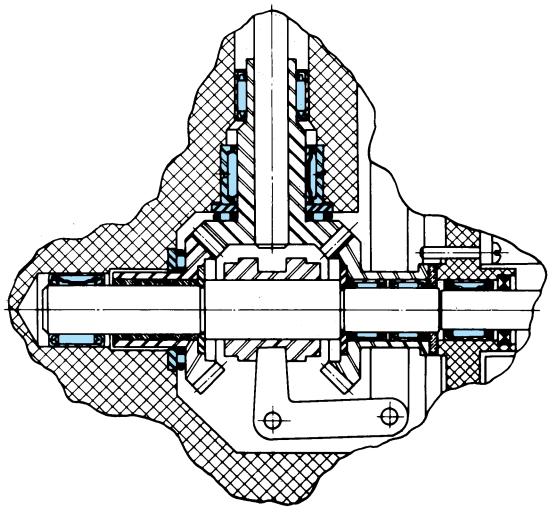


Fig. 81 Outboard Motor Lower Unit

HYDRAULIC MACHINES

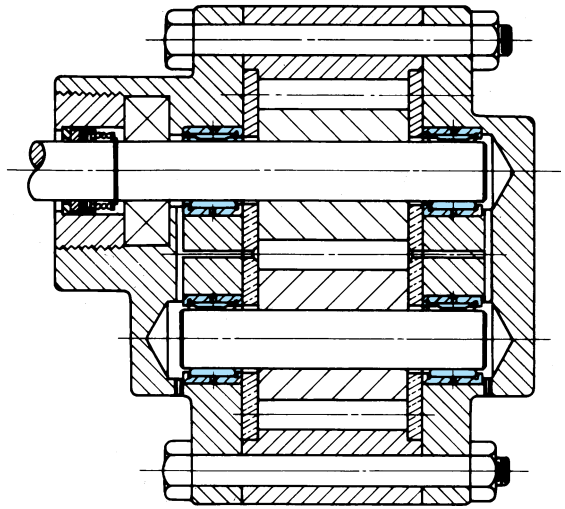


Fig. 82 Gear Pump
A solid needle roller bearing is commonly applied to high-pressure gear pumps. Since the load is large and problems such as edge load may be caused by shaft deflection, crowning should be applied to the rollers.

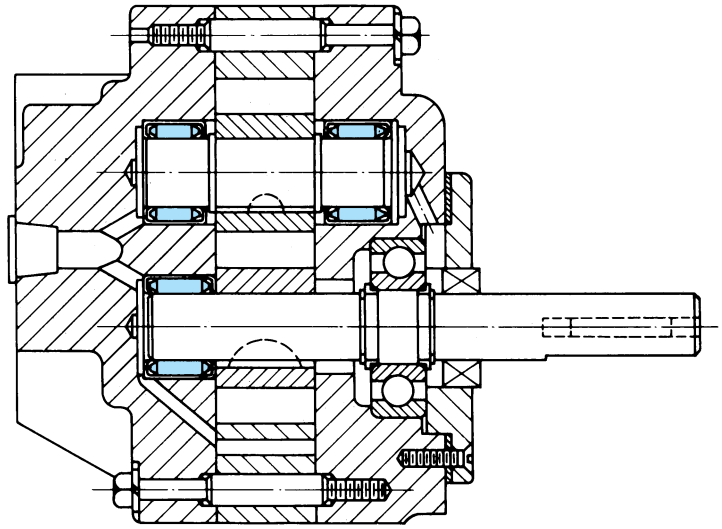


Fig. 83 Gear Pump
The comparatively low-pressure gear pump operates satisfactorily with a drawn cup needle roller bearing. It also serves to reduce cost.

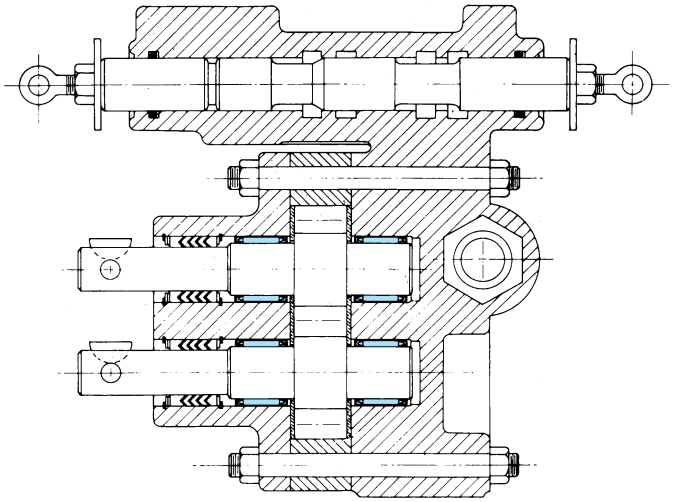


Fig. 84 Gear Pump

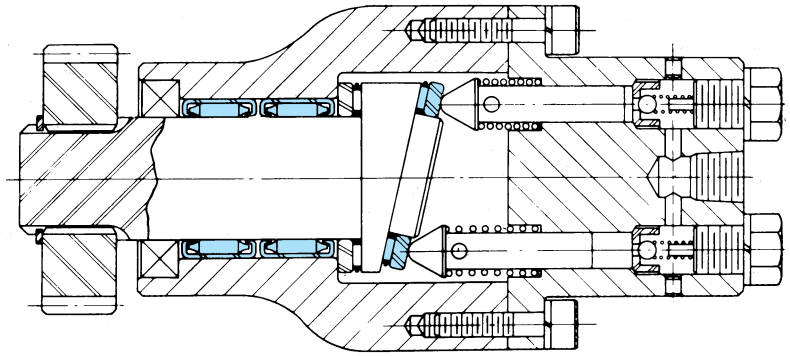


Fig. 85 Axial Piston Pump

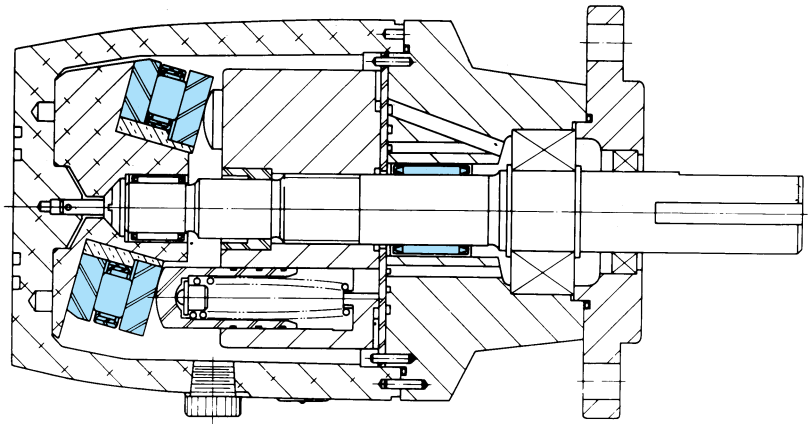


Fig. 86 Axial Piston Pump

For high-pressure axial piston pumps, NTH thrust roller bearings, which have a large load capacity, are recommended.
Although this example uses a copper alloy bush for the bore diameter guide face of the thrust roller bearing, a bush would not be required should the guide face be hardened and ground to HRC58 or more.

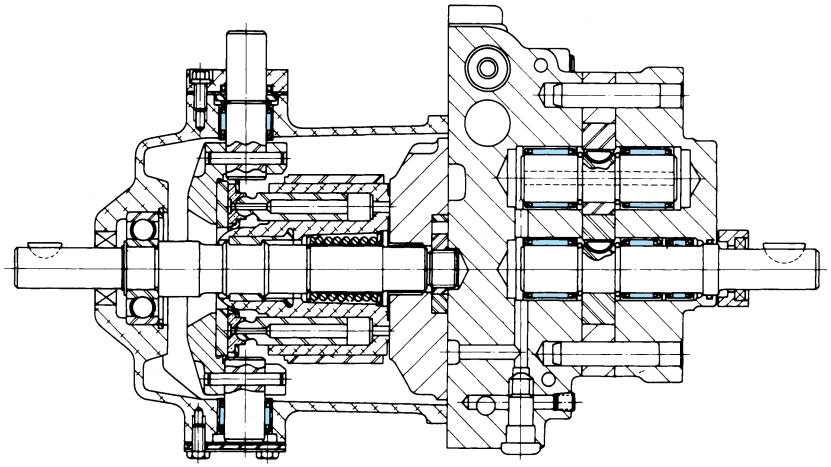


Fig. 87 Hydraulic Transmission

This illustrates a hydraulic transmission combining a variable displacement axial piston pump and a fixed displacement gear motor.
A drawn cup needle roller bearing is applied to the pivoting portion and the gear motor of the rotating swash plate of the axial piston pump.

OTHERS

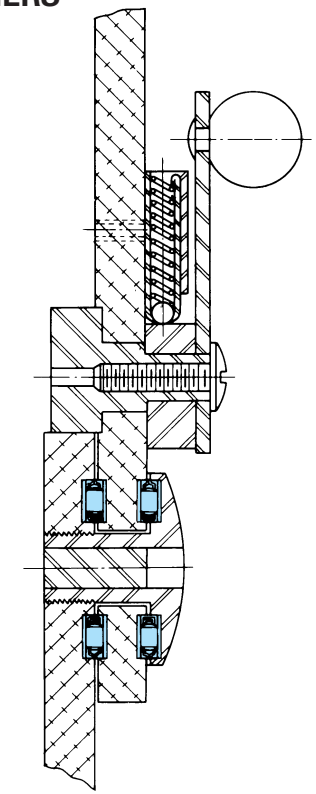


Fig. 88 Magic Hand
The application of a thrust needle roller bearing to a magic hand or artificial arm facilitates smooth operation.

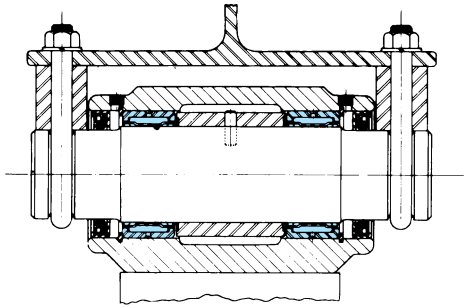


Fig. 89 Oil Well Equipment (Pump Jack)

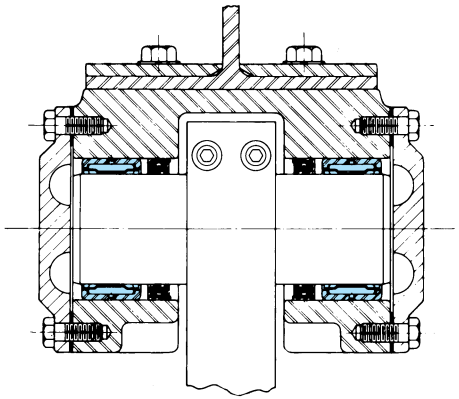


Fig. 90 Oil Well Equipment (Pump Jack)

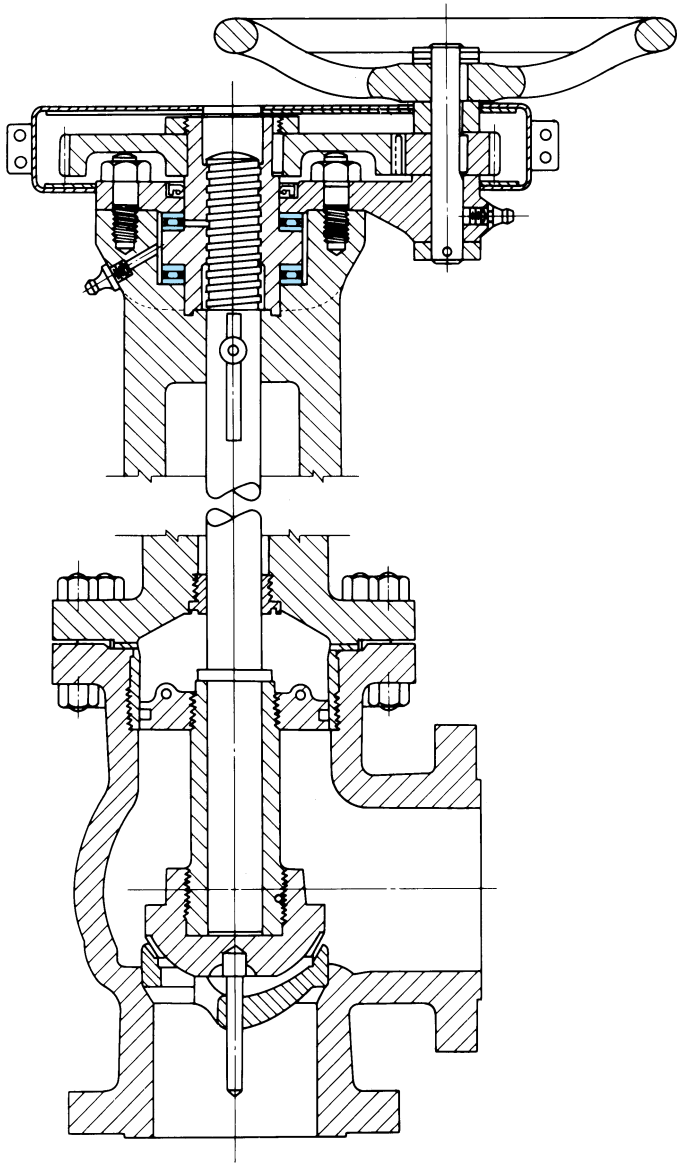


Fig. 91 Valve

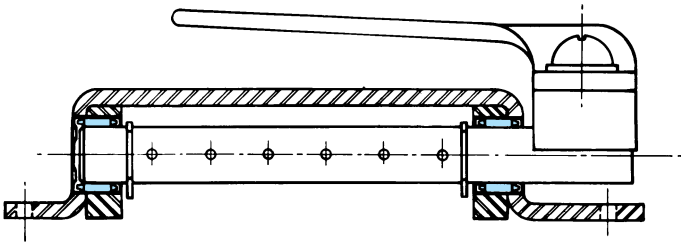


Fig. 92 Electric Guitar Tremolo

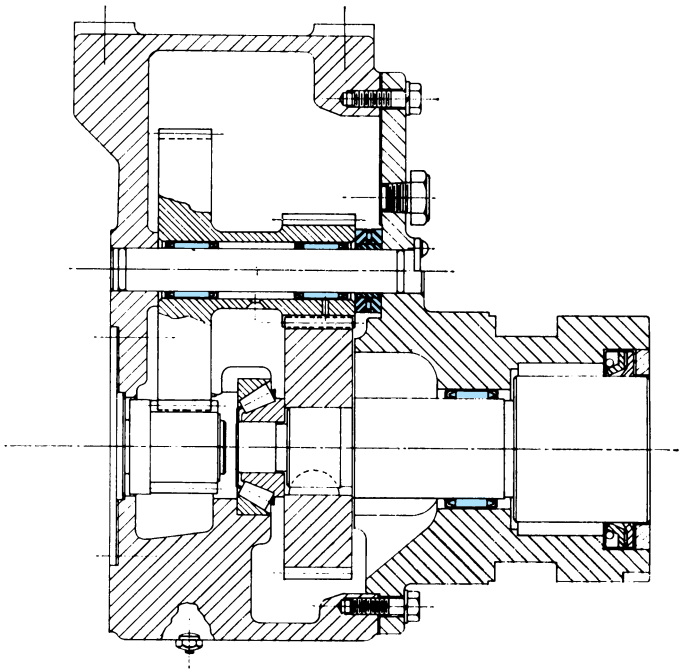


Fig. 93 Meat Saw for Hamburgers

Appendix 1 Conversion Table from SI (International Units) System

Comparison of SI, CGS, and Engineering Units

Units / Unit System	Length	Mass	Time	Temp.	Accelera-tion	Force	Stress	Pressure	Energy	Power
SI	m	kg	s	K, ℃	m/s^2	N	Pa	Pa	J	W
CGS System	cm	g	s	℃	Gal	dyn	dyn/cm^2	dyn/cm^2	erg	erg/s
Engineering Unit System	m	$kgf \cdot s^2/m$	s	℃	m/s^2	kgf	kgf/m^2	kgf/m^2	kgf·m	kgf·m/s

Conversion Factors from SI Units

Parameter	SI Units		Units other than SI		Conversion Factors from SI Units
	Names of Units	Symbols	Names of Units	Symbols	
Angle	Radian	rad	Degree Minute Second	° ' ''	$180/\pi$ $10\,800/\pi$ $648\,000/\pi$
Length	Meter	m	Micron Angstrom	μ Å	10^6 10^{10}
Area	Square meter	m^2	Are Hectare	a ha	10^{-2} 10^{-4}
Volume	Cubic meter	m^3	Liter Deciliter	l, L dl, dL	10^3 10^4
Time	Second	s	Minute Hour Day	min h d	1/60 1/3 600 1/86 400
Frequency	Hertz	Hz	Cycle	s^{-1}	1
Speed of Rotation	Revolution per second	s^{-1}	Revolution per minute	rpm	60
Speed	Meter per second	m/s	Kilometer per hour Knot	km/h kn	3 600/1 000 3 600/1 852
Acceleration	Meter per second per second	m/s^2	Gal g	Gal g	10^2 1/9.806 65
Mass	Kilogram	kg	Ton	t	10^{-3}
Force	Newton	N	Kilogram-force Ton-force Dyne	kgf tf dyn	1/9.806 65 $1/(9.806\,65 \times 10^3)$ 10^5
Torque or Moment	Newton·meter	N·m	Kilogram-force meter	kgf·m	1/9.806 65
Stress	Pascal	Pa (N/m^2)	Kilogram-force per square centimeter Kilogram-force per square milimeter	kgf/cm^2 kgf/mm^2	$1/(9.806\,65 \times 10^4)$ $1/(9.806\,65 \times 10^6)$

Prefixes Used In SI System

Multiples	Prefix	Symbols	Multiples	Prefix	Symbols
10^{18}	Exa	E	10^{-1}	Deci	d
10^{15}	Peta	P	10^{-2}	Centi	c
10^{12}	Tera	T	10^{-3}	Milli	m
10^9	Giga	G	10^{-6}	Micro	μ
10^6	Mega	M	10^{-9}	Nano	n
10^3	Kiro	k	10^{-12}	Pico	p
10^2	Hecto	h	10^{-15}	Femto	f
10	Deca	da	10^{-18}	Ato	a

Conversion Factors from SI Units (Continued)

Parameter	SI Units		Units other than SI		Conversion Factors from SI Units
	Names of Units	Symbols	Names of Units	Symbols	
Pressure	Pascal (Newton per square meter)	Pa (N/m^2)	Kilogram-force per square meter Water Column Mercury Column Torr Bar Atmosphere	kgf/m^2 mH_2O mmHg Torr bar atm	1/9.806 65 $1/(9.806\,65 \times 10^3)$ $760/(1.013\,25 \times 10^5)$ $760/(1.013\,25 \times 10^5)$ 10^{-5} $1/(1.013\,25 \times 10^5)$
Energy	Joule (Newton-meter)	J (N·m)	Erg Calorie (International) Kilogram-force meter Kilowatt hour French horse power hour	erg cal_{IT} kgf·m kW·h PS·h	10^7 1/4.186 8 1/9.806 65 $1/(3.6 \times 10^6)$ $\simeq 3.776\,72 \times 10^{-7}$
Work	Watt (Joule per second)	W (J/s)	Kilogram-force meter per second Kilocalorie per hour French horse power	kgf·m/s kcal/h PS	1/9.806 65 1/1.163 $\simeq$ 1/735.498 8
Viscosity, Viscosity Index	Pascal second	Pa·s	Poise	P	10
Kinematic Viscosity, Kinematic Viscosity Index	Square meter per second	m^2/s	Stokes Centistokes	St cSt	10^4 10^6
Temperature	Kelvin, Degree Celsius	K, ℃	Degree,	℃	(See Note (1))
Electric Current, Magnetomotive Force	Ampere	A	Ampere	A	1
Voltage, Electromotive Force	Volt	V	(Watts per ampere)	(W/A)	1
Magnetic Field Strength	Ampere per meter	A/m	Oersted	Oe	$4\pi/10^3$
Magnetic Flux Density	Tesla	T	Gauss Gamma	Gs γ	10^4 10^9
Electrical Resistance	Ohm	Ω	(Volts per ampere)	(V/A)	1

Note (1) The conversion from TK into θ ℃ is $\theta = T\text{-}273.15$ but for a temperature difference, it is $\Delta T = \Delta\theta$. However, ΔT and $\Delta\theta$ represent temperature differences measured using the Kelvin and Celsius scales respectively.

Remarks The names and symbols in ( ) are equivalent to those directly above them or on their left.
Example of conversion 1N = 1/9.806 65kgf

Appendix 2 N-kgf Conversion Table

(Method of using this table) For example, to convert 10N into kgf, read the figure in the right kgf column adjacent to the 10 in the center column in the 1st block. This means that 10N is 1.0197kgf. To convert 10kgf into N, read the figure in the left N column of the same row, which indicates that the answer is 98.066N.

1 N=0.1019716 kgf
1 kgf=9.80665 N

N		kgf	N		kgf	N		kgf
9.8066	1	0.1020	333.43	34	3.4670	657.05	67	6.8321
19.613	2	0.2039	343.23	35	3.5690	666.85	68	6.9341
29.420	3	0.3059	353.04	36	3.6710	676.66	69	7.0360
39.227	4	0.4079	362.85	37	3.7729	686.47	70	7.1380
49.033	5	0.5099	372.65	38	3.8749	696.27	71	7.2400
58.840	6	0.6118	382.46	39	3.9769	706.08	72	7.3420
68.647	7	0.7138	392.27	40	4.0789	715.89	73	7.4439
78.453	8	0.8158	402.07	41	4.1808	725.69	74	7.5459
88.260	9	0.9177	411.88	42	4.2828	735.50	75	7.6479
98.066	10	1.0197	421.69	43	4.3848	745.31	76	7.7498
107.87	11	1.1217	431.49	44	4.4868	755.11	77	7.8518
117.68	12	1.2237	441.30	45	4.5887	764.92	78	7.9538
127.49	13	1.3256	451.11	46	4.6907	774.73	79	8.0558
137.29	14	1.4276	460.91	47	4.7927	784.53	80	8.1577
147.10	15	1.5296	470.72	48	4.8946	794.34	81	8.2597
156.91	16	1.6315	480.53	49	4.9966	804.15	82	8.3617
166.71	17	1.7335	490.33	50	5.0986	813.95	83	8.4636
176.52	18	1.8355	500.14	51	5.2006	823.76	84	8.5656
186.33	19	1.9375	509.95	52	5.3025	833.57	85	8.6676
196.13	20	2.0394	519.75	53	5.4045	843.37	86	8.7696
205.94	21	2.1414	529.56	54	5.5065	853.18	87	8.8715
215.75	22	2.2434	539.37	55	5.6084	862.99	88	8.9735
225.55	23	2.3453	549.17	56	5.7104	872.79	89	9.0755
235.36	24	2.4473	558.98	57	5.8124	882.60	90	9.1774
245.17	25	2.5493	568.79	58	5.9144	892.41	91	9.2794
254.97	26	2.6513	578.59	59	6.0163	902.21	92	9.3814
264.78	27	2.7532	588.40	60	6.1183	912.02	93	9.4834
274.59	28	2.8552	598.21	61	6.2203	921.83	94	9.5853
284.39	29	2.9572	608.01	62	6.3222	931.63	95	9.6873
294.20	30	3.0591	617.82	63	6.4242	941.44	96	9.7893
304.01	31	3.1611	627.63	64	6.5262	951.25	97	9.8912
313.81	32	3.2631	637.43	65	6.6282	961.05	98	9.9932
323.62	33	3.3651	647.24	66	6.7301	970.86	99	10.095

Appendix 3 kg-lb Conversion Table

(Method of using this table) For example, to convert 10kg into lb, read the figure in the right lb column adjacent to the 10 in the center column in the 1st block. This means that 10kg is 22.046lb. To convert 10lb into kg, read the figure in the left kg column of the same row, which indicates that the answer is 4.536kg.

1 kg=2.2046226 lb
1 lb=0.45359237 kg

kg		lb	kg		lb	kg		lb
0.454	1	2.205	15.422	34	74.957	30.391	67	147.71
0.907	2	4.409	15.876	35	77.162	30.844	68	149.91
1.361	3	6.614	16.329	36	79.366	31.298	69	152.12
1.814	4	8.818	16.783	37	81.571	31.751	70	154.32
2.268	5	11.023	17.237	38	83.776	32.205	71	156.53
2.722	6	13.228	17.690	39	85.980	32.659	72	158.73
3.175	7	15.432	18.144	40	88.185	33.112	73	160.94
3.629	8	17.637	18.597	41	90.390	33.566	74	163.14
4.082	9	19.842	19.051	42	92.594	34.019	75	165.35
4.536	10	22.046	19.504	43	94.799	34.473	76	167.55
4.990	11	24.251	19.958	44	97.003	34.927	77	169.76
5.443	12	26.455	20.412	45	99.208	35.380	78	171.96
5.897	13	28.660	20.865	46	101.41	35.834	79	174.17
6.350	14	30.865	21.319	47	103.62	36.287	80	176.37
6.804	15	33.069	21.772	48	105.82	36.741	81	178.57
7.257	16	35.274	22.226	49	108.03	37.195	82	180.78
7.711	17	37.479	22.680	50	110.23	37.648	83	182.98
8.165	18	39.683	23.133	51	112.44	38.102	84	185.19
8.618	19	41.888	23.587	52	114.64	38.555	85	187.39
9.072	20	44.092	24.040	53	116.84	39.009	86	189.60
9.525	21	46.297	24.494	54	119.05	39.463	87	191.80
9.979	22	48.502	24.948	55	121.25	39.916	88	194.01
10.433	23	50.706	25.401	56	123.46	40.370	89	196.21
10.886	24	52.911	25.855	57	125.66	40.823	90	198.42
11.340	25	55.116	26.308	58	127.87	41.277	91	200.62
11.793	26	57.320	26.762	59	130.07	41.730	92	202.83
12.247	27	59.525	27.216	60	132.28	42.184	93	205.03
12.701	28	61.729	27.669	61	134.48	42.638	94	207.23
13.154	29	63.934	28.123	62	136.69	43.091	95	209.44
13.608	30	66.139	28.576	63	138.89	43.545	96	211.64
14.061	31	68.343	29.030	64	141.10	43.998	97	213.85
14.515	32	70.548	29.484	65	143.30	44.452	98	216.05
14.969	33	72.753	29.937	66	145.51	44.906	99	218.26

Appendix 4 ℃-℉ Conversion Table

(Method of using this table) For example, to convert 38℃ into ℉, read the figure in the right ℉ column adjacent to the 38 in the center column in the 2nd block. This means that 38℃ is 100.4℉. To convert 38℉ into ℃, read the figure in the left ℃ column of the same row, which indicates that the answer is 3.3℃.

$$C=\frac{5}{9}(F-32)$$

$$F=32+\frac{9}{5}C$$

℃		℉	℃		℉	℃		℉	℃		℉
−73.3	**−100**	−148.0	0.0	**32**	89.6	21.7	**71**	159.8	43.3	**110**	230
−62.2	**− 80**	−112.0	0.6	**33**	91.4	22.2	**72**	161.6	46.1	**115**	239
−51.1	**− 60**	− 76.0	1.1	**34**	93.2	22.8	**73**	163.4	48.9	**120**	248
−40.0	**− 40**	− 40.0	1.7	**35**	95.0	23.3	**74**	165.2	51.7	**125**	257
−34.4	**− 30**	− 22.0	2.2	**36**	96.8	23.9	**75**	167.0	54.4	**130**	266
−28.9	**− 20**	− 4.0	2.8	**37**	98.6	24.4	**76**	168.8	57.2	**135**	275
−23.3	**− 10**	14.0	3.3	**38**	100.4	25.0	**77**	170.6	60.0	**140**	284
−17.8	**0**	32.0	3.9	**39**	102.2	25.6	**78**	172.4	65.6	**150**	302
−17.2	**1**	33.8	4.4	**40**	104.0	26.1	**79**	174.2	71.1	**160**	320
−16.7	**2**	35.6	5.0	**41**	105.8	26.7	**80**	176.0	76.7	**170**	338
−16.1	**3**	37.4	5.6	**42**	107.6	27.2	**81**	177.8	82.2	**180**	356
−15.6	**4**	39.2	6.1	**43**	109.4	27.8	**82**	179.6	87.8	**190**	374
−15.0	**5**	41.0	6.7	**44**	111.2	28.3	**83**	181.4	93.3	**200**	392
−14.4	**6**	42.8	7.2	**45**	113.0	28.9	**84**	183.2	98.9	**210**	410
−13.9	**7**	44.6	7.8	**46**	114.8	29.4	**85**	185.0	104.4	**220**	428
−13.3	**8**	46.4	8.3	**47**	116.6	30.0	**86**	186.8	110.0	**230**	446
−12.8	**9**	48.2	8.9	**48**	118.4	30.6	**87**	188.6	115.6	**240**	464
−12.2	**10**	50.0	9.4	**49**	120.2	31.1	**88**	190.4	121.1	**250**	482
−11.7	**11**	51.8	10.0	**50**	122.0	31.7	**89**	192.2	148.9	**300**	572
−11.1	**12**	53.6	10.6	**51**	123.8	32.2	**90**	194.0	176.7	**350**	662
−10.6	**13**	55.4	11.1	**52**	125.6	32.8	**91**	195.8	204	**400**	752
−10.0	**14**	57.2	11.7	**53**	127.4	33.3	**92**	197.6	232	**450**	842
− 9.4	**15**	59.0	12.2	**54**	129.2	33.9	**93**	199.4	260	**500**	932
− 8.9	**16**	60.8	12.8	**55**	131.0	34.4	**94**	201.2	288	**550**	1022
− 8.3	**17**	62.6	13.3	**56**	132.8	35.0	**95**	203.0	316	**600**	1112
− 7.8	**18**	64.4	13.9	**57**	134.6	35.6	**96**	204.8	343	**650**	1202
− 7.2	**19**	66.2	14.4	**58**	136.4	36.1	**97**	206.6	371	**700**	1292
− 6.7	**20**	68.0	15.0	**59**	138.2	36.7	**98**	208.4	399	**750**	1382
− 6.1	**21**	69.8	15.6	**60**	140.0	37.2	**99**	210.2	427	**800**	1472
− 5.6	**22**	71.6	16.1	**61**	141.8	37.8	**100**	212.0	454	**850**	1562
− 5.0	**23**	73.4	16.7	**62**	143.6	38.3	**101**	213.8	482	**900**	1652
− 4.4	**24**	75.2	17.2	**63**	145.4	38.9	**102**	215.6	510	**950**	1742
− 3.9	**25**	77.0	17.8	**64**	147.2	39.4	**103**	217.4	538	**1 000**	1832
− 3.3	**26**	78.8	18.3	**65**	149.0	40.0	**104**	219.2	593	**1 100**	2012
− 2.8	**27**	80.6	18.9	**66**	150.8	40.6	**105**	221.0	649	**1 200**	2192
− 2.2	**28**	82.4	19.4	**67**	152.6	41.1	**106**	222.8	704	**1 300**	2372
− 1.7	**29**	84.2	20.0	**68**	154.4	41.7	**107**	224.6	760	**1 400**	2552
− 1.1	**30**	86.0	20.6	**69**	156.2	42.2	**108**	226.4	816	**1 500**	2732
− 0.6	**31**	87.8	21.1	**70**	158.0	42.8	**109**	228.2	871	**1 600**	2912

Appendix 5 Viscosity Conversion Table

Kinematic Viscosity mm²/s	Saybolt Universal SUS (sec)		No.1 Type Redwood R (sec)		Engler E (degree)
	100℉	210℉	50℃	100℃	
2	32.6	32.8	30.8	31.2	1.14
3	36.0	36.3	33.3	33.7	1.22
4	39.1	39.4	35.9	36.5	1.31
5	42.3	42.6	38.5	39.1	1.40
6	45.5	45.8	41.1	41.7	1.48
7	48.7	49.0	43.7	44.3	1.56
8	52.0	52.4	46.3	47.0	1.65
9	55.4	55.8	49.1	50.0	1.75
10	58.8	59.2	52.1	52.9	1.84
11	62.3	62.7	55.1	56.0	1.93
12	65.9	66.4	58.2	59.1	2.02
13	69.6	70.1	61.4	62.3	2.12
14	73.4	73.9	64.7	65.6	2.22
15	77.2	77.7	68.0	69.1	2.32
16	81.1	81.7	71.5	72.6	2.43
17	85.1	85.7	75.0	76.1	2.54
18	89.2	89.8	78.6	79.7	2.64
19	93.3	94.0	82.1	83.6	2.76
20	97.5	98.2	85.8	87.4	2.87
21	102	102	89.5	91.3	2.98
22	106	107	93.3	95.1	3.10
23	110	111	97.1	98.9	3.22
24	115	115	101	103	3.34
25	119	120	105	107	3.46
26	123	124	109	111	3.58
27	128	129	112	115	3.70
28	132	133	116	119	3.82
29	137	138	120	123	3.95
30	141	142	124	127	4.07
31	145	146	128	131	4.20
32	150	150	132	135	4.32
33	154	155	136	139	4.45
34	159	160	140	143	4.57

Kinematic Viscosity mm²/s	Saybolt Universal SUS (sec)		No.1 Type Redwood R (sec)		Engler E (degree)
	100℉	210℉	50℃	100℃	
35	163	164	144	147	4.70
36	168	170	148	151	4.83
37	172	173	153	155	4.96
38	177	178	156	159	5.08
39	181	183	160	164	5.21
40	186	187	164	168	5.34
41	190	192	168	172	5.47
42	195	196	172	176	5.59
43	199	201	176	180	5.72
44	204	205	180	185	5.85
45	208	210	184	189	5.98
46	213	215	188	193	6.11
47	218	219	193	197	6.24
48	222	224	197	202	6.37
49	227	228	201	206	6.50
50	231	233	205	210	6.63
55	254	256	225	231	7.24
60	277	279	245	252	7.90
65	300	302	266	273	8.55
70	323	326	286	294	9.21
75	346	349	306	315	9.89
80	371	373	326	336	10.5
85	394	397	347	357	11.2
90	417	420	367	378	11.8
95	440	443	387	399	12.5
100	464	467	408	420	13.2
120	556	560	490	504	15.8
140	649	653	571	588	18.4
160	742	747	653	672	21.1
180	834	840	734	757	23.7
200	927	933	816	841	26.3
250	1 159	1 167	1 020	1 051	32.9
300	1 391	1 400	1 224	1 241	39.5

Remarks $1mm^2/s=1cSt$